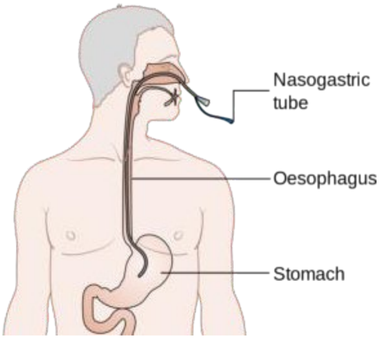
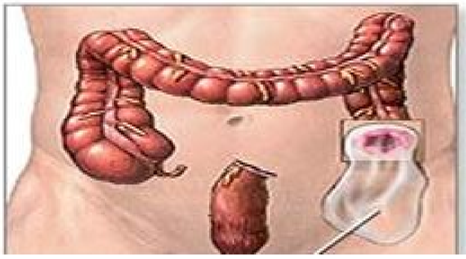
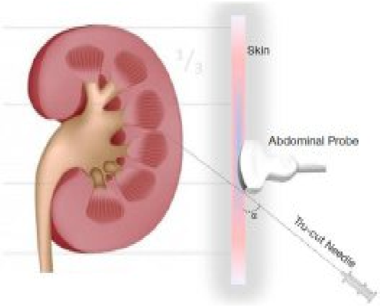
Medical Surgical Nursing Clinical Procedures
Second Term







Prepared by

Medical Surgical Nursing Department 2025/2026

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Faculty of Nursing

Vision and Mission 2025-2026

Vision:-

To address health and technical concerns at the local, national, and regional levels as well as hygiene, the medical surgical nursing department will establish an innovative academic Centre for education, practice, research, and community services.

Mission :-

To develop a capable student who can compete at the local, national, and regional levels and who is motivated to learn new things and do scientific research. The student should also be qualified to provide comprehensive nursing care for a variety of medical surgical nursing specialties.

Target

1. Raising the efficiency of the institutional capacity to improve the competitive position of the college.
2. Excellence in teaching and learning to increase the competitiveness of college students.
3. Developing and supporting scientific research and focusing it on community service.
4. Expansion of community service and environmental development.

Strategic Plan

The Medical Surgical Nursing Department strategic plan is a living document that guides the work of RST. The purpose of the plan is to focus all RST initiatives on achieving RST mission of commitment to quality patient care through professional development, certification, scholarship, and advocacy.

Goal I:

RST Nurse will be a well-equipped, highly respected professional with career satisfaction and a strong professional identity.
Objectives:

1. Increase the competency and capability of the medical-surgical nurse at all levels of their career.
2. Increase career satisfaction among medical-surgical nurses.
3. Increase the professional identity of med-surg nurses.

Goal 2:

The global healthcare community will look to RST as the thought leader on all aspects of med-surg nursing practice through advocacy, education, certification, and scholarship.
Objectives:

1. Increase acknowledgement of med-surg nursing as a specialty.
2. Increase the profile of the med-surg nurse as a healthcare leader.
3. Increase recognition of RST as an education authority on topics related to med-surg nursing.
4. Increase promotion of the value of certification as a hallmark of expertise.
5. Increase scholarship on issues relevant to med-surg nursing through research activities.

Goal 3:

RST shapes the field of medical-surgical nursing through a diverse workforce and equitable opportunities for all nurses.
Objectives:

1. Demonstrate a commitment to increasing diversity, inclusion, and equity within the organization.

2. Increase medical-surgical nursing's focus on SDOH (social determinants of health) and implement strategies to promote health equity.
3. Increase competence in med-surg nurses to improve patient care and practice environments.
4. Increase med-surg nurses' ability to lead in their communities.
5. Increase med-surg nurses' ability to apply evidence-based practice and research, in their day-to-day practices for better patient outcomes.

Goal 4:

RST health and continued organizational growth will be advanced through the effective integration of products, programs, technology, systems, and people.

Objectives:

1. Improve member experience and engagement.
2. Engage current membership to increase volunteer capabilities, opportunities, and leadership succession.
3. Develop and implement an effective governance structure and associated processes.

Responsibilities of Students in Clinical Lab: -

- Review the skills to be practiced and/or demonstrated as well as having read the assigned chapters by your instructor prior to lab attendance.
- Gather and return the equipment used for skill performance.
- Inform instructor when unable to locate equipment, supplies or resources.
- Handle simulators and equipment with care and respect.
- Follow safety measures at all times.
- Maintain cleanliness of the area.
- Dispose of sharps appropriately.

- Demonstrate respect and consideration for self and others.
- All students should display professional conduct.
- Allow others the opportunity to practice.
- Inform instructor if supplies are running low.
- Any damage or malfunction of mannequins or equipment should be reported to the lab faculty immediately.

Course Content

1. Demonstrate procedure of colostomy care and assist in liver, renal biopsy technique
2. Identify principles of aseptic technique
3. Identify types of liver and renal biopsy.
4. Demonstrate procedure of dressing technique, how to prepare package and dressing care regarding burn wound
5. Identify types of cast and traction
6. Define hydrotherapy & debridement
7. Perform nursing care in endoscopic procedure
8. Identify the nursing implication in medical procedures as liver and renal biopsy

1. <u>Topics outlines:</u>

*Gastrointestinal Tract

●Nasogastric Tube insertion and care" gavage

●Colostomy Care and Colostomy Irrigation

●Liver Biopsy

●Paracentesis

●Endoscopy procedure

*Burn wound care

- Wound hydrotherapy
- Wound debridement
- Wound dressing

*Orthopedic procedures

- Cast Procedure
- Traction Procedure

*Neurological procedures

- Lumbar puncture

*Renal Procedure

- Renal Biopsy

2.Teaching strategies:

1- Audiovisual aids: - Models, Equipment.

2- Demonstration.

3- Guided practice in laboratory and clinical/hospital settings.

2. Videos

3.Evaluation methods:

1- Clinical evaluation

2- Case studies.

The final grade in the course is based on the following.

4.Case study: - two case studies are required per semester, one of which should be presented by the student during a class or clinical conference.

5.Final clinical assessment:

1- The final clinical examination should be done in the last two weeks of the semester.

2- The case should be chosen at the beginning of the examination day.

3- Cases selected for the examination should be from medical and surgical units.

Module Unit: - Elimination

Module Title: - Colostomy Care and Colostomy Irrigation Time

Allocated: - one hour theory / 30 Minute clinical practice

(Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with colostomy.
  - Implements standardized protocol and guidelines when providing nursing care for patients with colostomy.
  - Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with colostomy.
  - Describe the responsibilities of the nurse in meeting the psychological needs of the patients with colostomy.
  - Develop a teaching plan for patients with colostomy.
  - Manage time effectively and set priorities.
  - Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
  - Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
  - Convey a positive attitude toward other team members while working with patients with colostomy.

Teaching and Learning Activities:

a) Oral Presentation.

b) Group participation

c) Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Outlines:

●Definition of colostomy& colostomy irrigation.

- Purposes of colostomy care and colostomy irrigation.
- Equipment needed for colostomy care and colostomy irrigation.
- Steps of colostomy care procedure and colostomy irrigation
- Identify Characteristics of healthy Stoma.
- Explain complications of colostomy.

Key terms: -

1) Colostomy: - The surgically created opening of the colon (large intestine) which results in a stoma.
2) A temporary colostomy (usually transverse) may be raised to divert the fecal output, thus allowing healing of an anastomosis further along the colon.
3) Sigmoid or Descending Colostomy: - The most common type of ostomy surgery, in which the end of the descending or sigmoid colon is brought to the surface of the abdomen.
4) Transverse colostomy: -

The surgical opening created in the transverse colon resulting in one or two openings.

5) Ascending Colostomy: - A relatively rare opening in the ascending portion of the colon. It is located on the right side of the abdomen.
6) Ileoanal anastomosis: - Technically, it is not an ostomy since there is no stoma. In this procedure, the colon and most of the rectum are surgically removed and an internal pouch is formed out of the terminal portion of the ileum.
7) Colostomy Pouches: - Open-ended pouches are called drainable and are left attached to the body while emptying. Most commonly closed end pouches are used by colostomies.
8) Irrigation Systems: - This is done to clean stool directly out of the

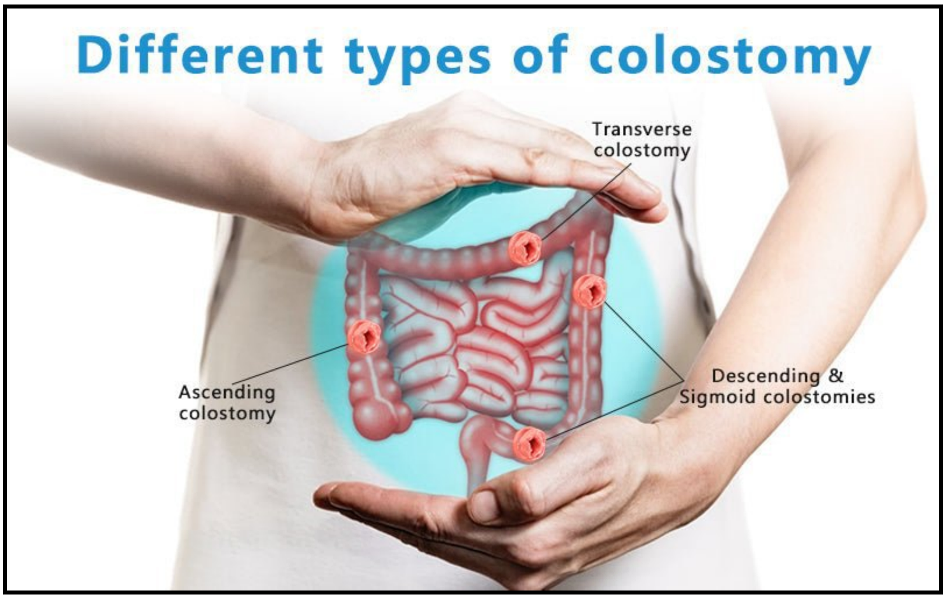
colon through the stoma

Indications of colostomy:

- A blockage
- An injury
- Crohn's disease, which is an autoimmune form of inflammatory bowel disease
- Colorectal cancer
- Colonic polyps, which is extra tissue growing inside the colon that may be cancer or may turn into cancer
- Diverticulitis, which occurs when small pouches in digestive system, called diverticula, become infected or inflamed
- Imperforate anus or other birth defects
- Irritable bowel syndrome, which is a condition affecting the colon that causes diarrhea, bloating, constipation, and pain in the abdominal area
- Ulcerative colitis, which is an inflammatory bowel disease that causes the long-term inflammation of the digestive tract.

Types of colostomy

1-according to site



Ascending Colostomy

The ascending colostomy is located on the right side of the abdomen. The discharge is very liquid. A drainable pouch is worn for colostomies of this type. This type of stoma is rarely used since an ileostomy is a better stoma when the discharge is liquid. When a colostomy is located in the right half of the colon, only a short portion of colon remains.

Transverse Colostomy

The transverse colostomy is in the upper abdomen, either in the middle or toward the right side of the body. This type of colostomy allows stools to exit the colon before it reaches the descending colon.

A transverse colostomy may be created for a period of time to prevent stool from passing through the area of the colon that is inflamed, infected, diseased or newly operated on, thus allowing healing to occur.

Such a colostomy is usually temporary. Depending on the healing process, colostomy may be necessary for a few weeks, months, or even years. Eventually given your good health, the colostomy is likely to be closed and normal bowel continuity restored.

The discharge from the transverse colostomy is semi-solid, unpredictable and contains some digestive enzymes. The management of the transverse colostomy consists of skin protection and a drainable pouch. A closed-end pouch can be used for convenience during special activities.

Descending colostomy

Located on the lower left side of the abdomen. Generally, the discharge is firm and can be regulated. The sigmoid colostomy is probably the most frequently performed of all the colostomies.

The stool of a descending or sigmoid colostomy is firmer than that of the transverse colostomy and does not have the caustic enzyme content. At this location, elimination may occur on a reflex basis at regular, predictable intervals.

2-According to time

1. Permanent colostomy.

Permanent colostomy	Temporary colostomy
A permanent it provides a mean of elimination when the end portion of colon, rectum or anus in nonfunctional and must be totally removed.	It is created for elimination when healing needs to take place in the case of trauma or inflammatory condition of the bowel.
It used for long term and may be for long life.	It used for few weeks, months or even years.
It will not be closed at any time.	It will be closed and normal bowel continuity is restored.

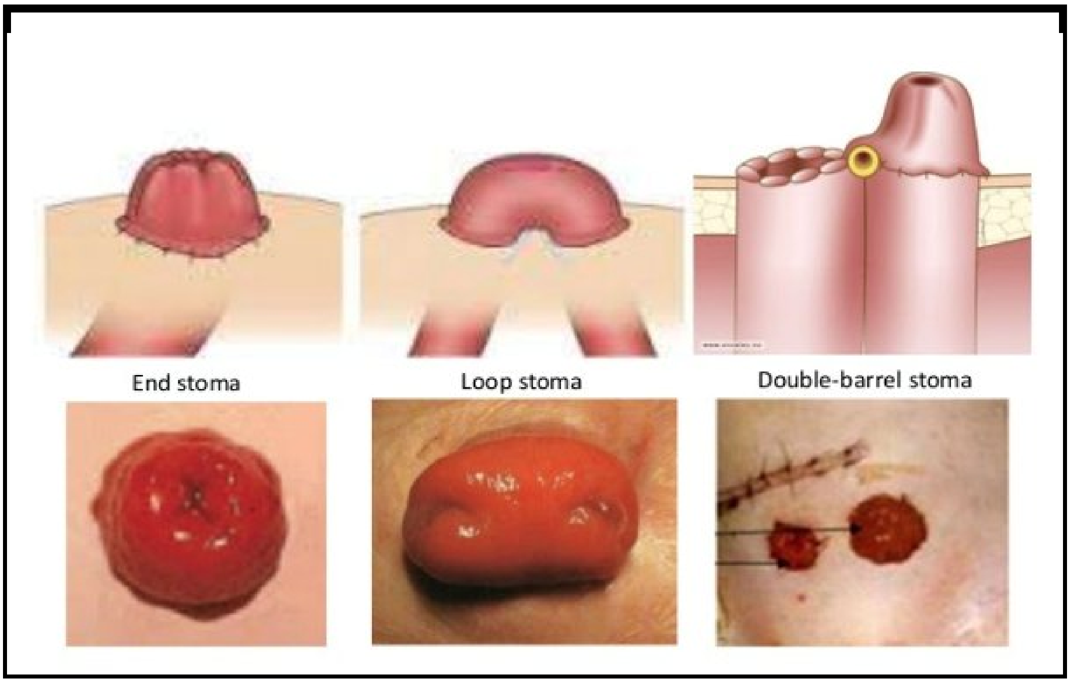
2. Temporary colostomy.

Types of stomas

There are three types of stomas that can be created for a colostomy: “loop stoma,” “end stoma,” and “loop-end stoma.”

1- The “end stoma” is one stoma with one opening. To create an “end stoma” the surgeon brings the open end of the bowel out through a tiny incision in the abdominal wall, folds it over similar to a cuff and sutures it down.

2- A loop stoma is one stoma with two different openings. An entire loop of bowel is brought out through the abdominal surface through one incision and then opened exposing two pathways into the bowel: a proximal (functional) opening and a distal (nonfunctional) opening. The proximal opening expels stool as it travels down through the bowel and the distal opening allows mucous to exit the distal or lower portion of the bowel.



3-A double barrel ostomy is a surgical procedure which creates two end stomas usually in close proximity. The proximal stoma is a functional opening which expels stool and the distal stoma functions simply as a mucous fistula.

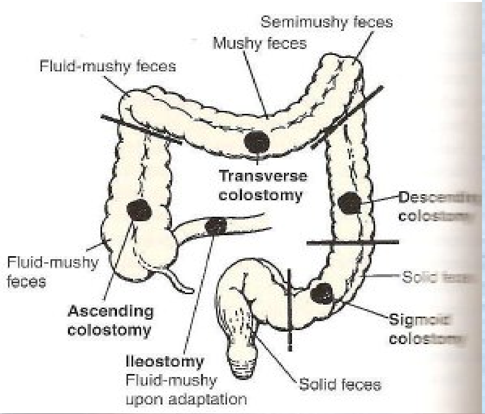
Colostomy Care

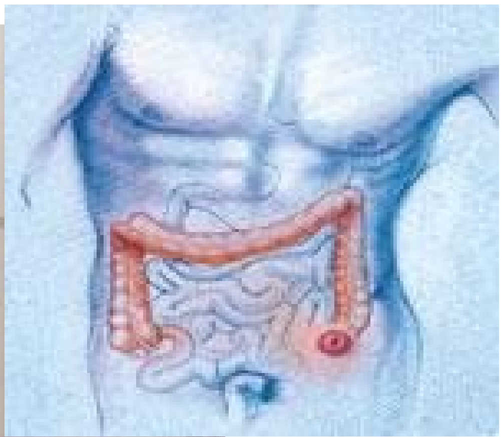
Types of fecal ostomy:

1. Colostomy:

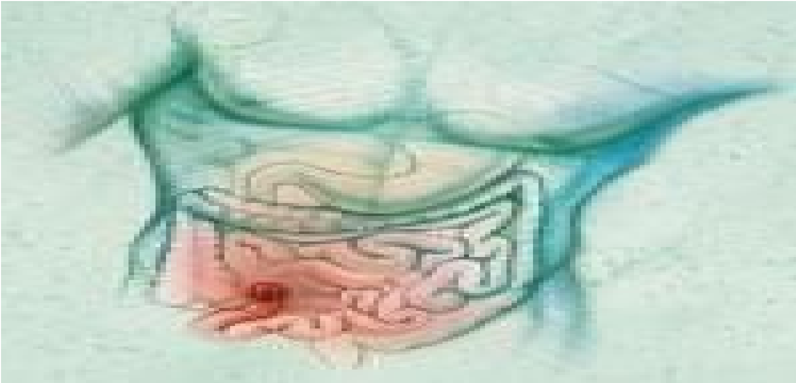
A surgically created opening between the colon & the abdominal wall to allow fecal elimination. It may be temporary or permanent diversion.

Colostomy may be placed in any segment of large intestine (colon), which will influence the nature of fecal discharge. The more right sided the colostomy the looser the stool. Transverse and descending/sigmoid colostomies are the most common types.






2. Ileostomy:



A surgically created opening between the ileum of the small intestine and abdominal wall to allow elimination of small bowel effluent. An ileostomy is usually formed at the terminal ileum of the small bowel and is usually placed in the right lower quadrant of the abdomen. Stool from an ileostomy drains frequently (average 4 to 5 times per day) and contains proteolytic enzymes, which can be harmful to skin.

Purpose of colostomy care:

1. To provide an outlet for intestinal waste product (drainage of fecal matter).
2. To maintains integrity of stoma and peristomal skin (skin surrounding stoma)
3. To prevents lesions, ulcerations, excoriation, and other skin breakdown caused by fecal contaminants
4. To prevents infection
5. To promotes general comfort and positive self-image /self-concept
6. To provides clean ostomy pouch for fecal evacuation
7. To reduces odor from overuse of old pouch.

Equipment:

1- Duplicate wafer and pouch.

2- Washcloth and towel.

3- Tail closure

4- Mild non oily soap (optional)

5- Accessory products prescribed for patient.

Procedure:

Nursing Action	Rationale
Preparatory phase:	
Have patient assume a relaxed position and provide privacy. The best position may be sitting, reclining, or standing.	Patient must see stoma site to learn care.
Performance phase:	
1. To remove pouching system:	
a. Wear nonsterile gloves.	Maintain universal precautions.
b. Push down gently on skin while lifting up on the wafer (ostomy adhesive remover may be used).	b. Minimizes skin trauma.
c. Discard soiled pouch and wafer in odor proof plastic bag. Save tail closure for reuse.	c. Removes room odor and maintain universal precaution.
2. To cleans skin:	2.
a. Use toilet tissue to remove feces from stoma and skin if needed.	Stoma may function during the change.
b. Cleans stoma and peristomal	b. Minimize skin break down and promotes hygiene.

with or without pouching system in place. Clip or shave peristomal hair if appropriate.	
c. Rinse and dry skin thoroughly after cleansing. It is normal for stoma to bleed slightly during cleaning and drying.	c. Remove residue, which may interfere with adhesion of wafer.
3. To apply wafer:	
a. Use measuring guide or pattern to determine stoma size.	a. This step is omitted when stomal shrinkage is complete, about 2 months postoperatively.
b. Trace correct size onto back of wafer and cut to stoma size. It is acceptable to cut 1/16- 1/8 inch larger than stoma.	b. Avoids wafer rubbing stoma; omit this step if the wafer is precut
c. Apply a line of skin barrier paste around stoma or on tip of wafer opening. Allow to set according to manufacturer's instructions.	c. Extra skin protection is imperative for ileostomy and right sided colostomy. A left sided colostomy may not need a secondary barrier because formed stool is less harmful to skin. Paste acts as caulking to prevent undermining of feces.
d. Remove paper backing from the wafer, center opening over stoma, and press wafer down onto peristomal skin.	d. Ensures adherence.
4. Snap pouch on to the flange of water according to	4. If attached properly, there will be no leakage or odor.

directions.

5. Apply tail closure to pouch tail — 5. Proper closure controls odor.

Follow up phase:

1. Dispose of plastic bag with waste materials.
2. Clean drainable pouch with soap and water; if appropriate. Drainable pouches may be reused several times

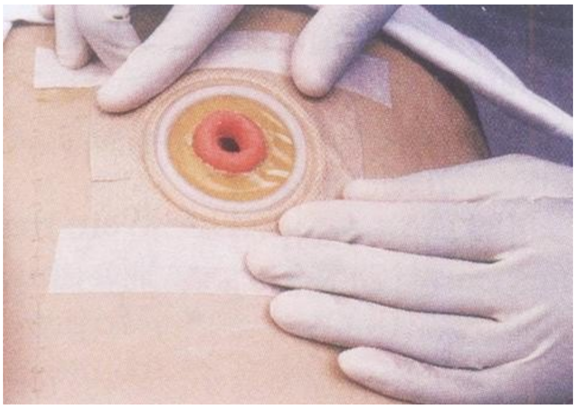
Complies with universal precautions. Controls odor: reduces cost.

3. A commercial deodorant can be placed in the pouch to reduce odor.
4. Gas can be released from the pouch by releasing the tail closure or by snapping off an area on the pouch flange. Never make a pinhole in the pouch to release gas.

Destroys the odor proof seal.

Characteristics of healthy Stoma:

1. Stoma is part of intestine (small or large) that is brought above the abdominal wall to become the outlet for discharge of intestinal waste.
2. Pink –red.
3. Moist.
4. Bleeds slightly when rubbed.
5. No feeling of touch.



6. Stool functions involuntary.

7. Postoperative swellings gradually decrease over several months.

Complications:

1.Mucocutaneous separation (skin & stoma)

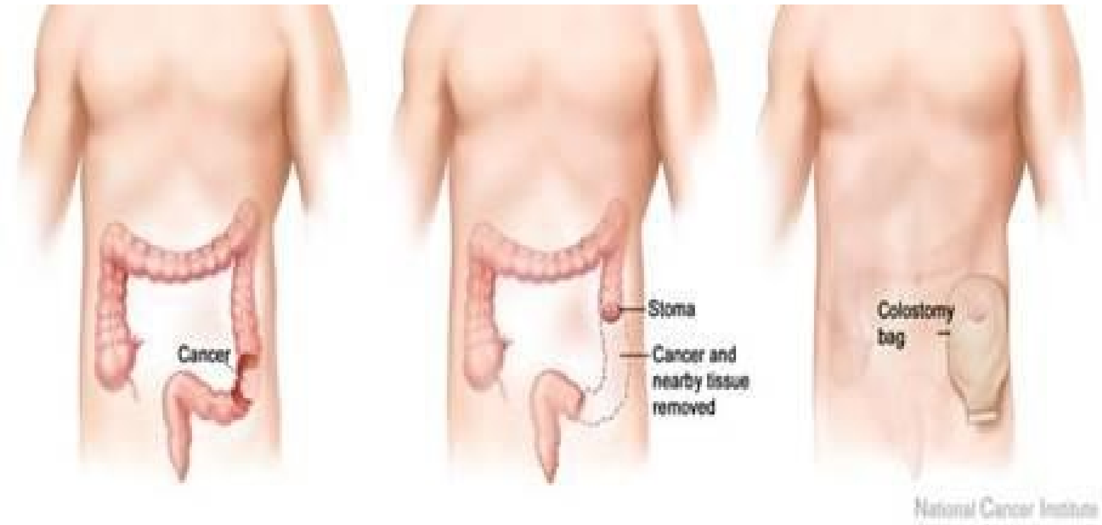
2.Stomal ischemia

3.Stomal stricture or stenosis (long term)

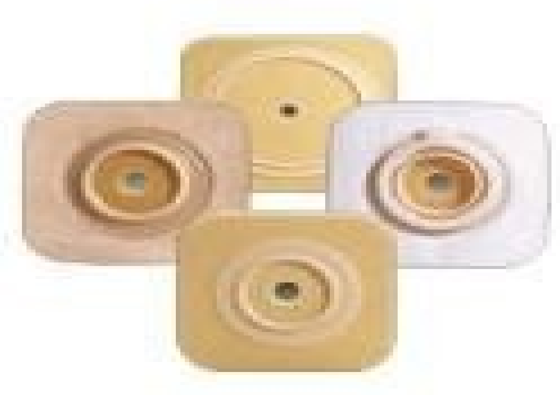
4.Stomal prolapse

5.Peristomal hernia

6.Peristomal skin break down (exposure to fecal output, allergic reaction, infection).



Procedure



Appliance

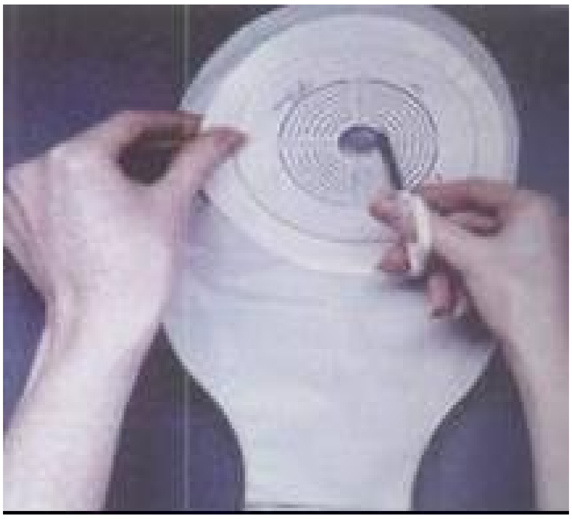
Pouches

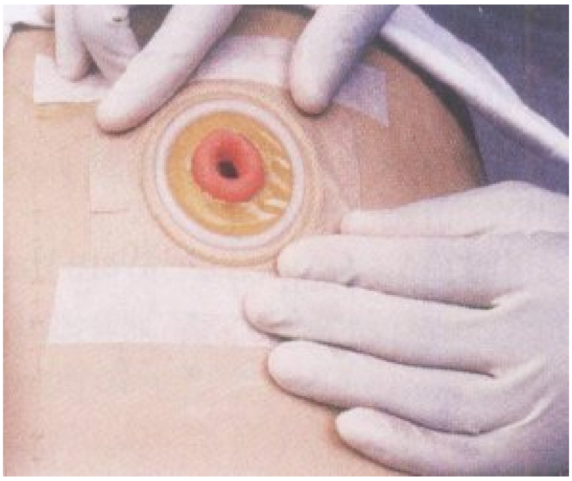
Stoma Measuring Guide

Removing Appliance

Cleaning Stoma

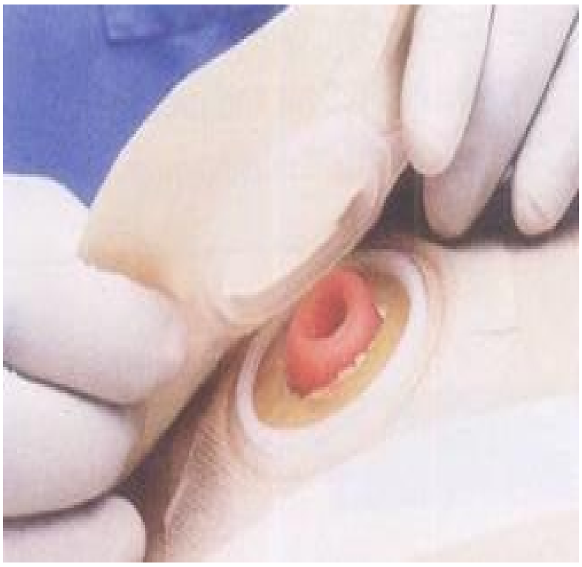
Measuring Stoma





past the appliance

Cutting Wafer



Colostomy irrigation

A procedure in which a patient with a colostomy flushes the colon with water, using a tube that is inserted into the stoma (a surgically created opening in the body that connects an organ or area inside the body with the outside). This causes the colon to empty and pass stool through the stoma into a bag. The procedure should be done at the same time every day. It may allow colostomy patients to have better control over their bodies.

Purpose:

- To empty the colon of feces, gas, mucus.
- To establish a regular pattern of evacuation and prevent constipation.

Equipment:

- Irrigating sleeve with belt.
- Water-soluble lubricant.
- Plastic garbage bag.
- Appliance.
- Bedpan or large bowl if the patient is bedbound.

- Soap and warm water, basin, washcloths, and towels.
- Disposable non-sterile gloves, disposable apron.

Procedure:

Step

1. Explain to procedure to the patient/caregiver.
2. Assist the patient to sit on the toilet or in a chair facing the toilet.
3. Fill the irrigation bag with 500 to 1000 ml of water.
4. Flush the tubing with water to empty if of air.
5. Hang up the irrigation bag on the bathroom door hook, or raise the bag 18 inches above the stoma before irrigation.
6. Take off the pouch and the soiled colostomy bag.
7. Remove any stool or drainage from the stoma with toilet paper.
8. Gently clean with soap and warm water as needed.
9. Place the ring and irrigation sleeve over the stoma and adjust the belt for a snug fit (a mild soapy solution may be squirted into the sleeve for easier cleaning).
10. Put the bottom of the sleeve into the toilet.
11. Attach the cone tip to the irrigation tubing and lubricate.
12. Gently insert the cone into the stoma for a snug fit. If resistance is met, perform a lubricated digital exam to remove feces. NEVER use force because this could perforate the bowel.

13. Open the clamp and let the water flow in over 15 minutes. Press on the cone just firmly enough to keep water from running back out. If cramps occur, turn off the water with the clamp and wait until the cramps subside before proceeding. If the water backs up into the tubing, shut off the water with the clamp. Wait a few minutes, then open the clamp and continue irrigation (lower the bag to decrease the flow and raise the bag to increase the flow of water).
14. Close the bottom of the sleeve with the clamp when most of the water has come back out (about 10 to 20 minutes).
15. Instruct the patient to walk around if able, for about 30 minutes to complete the emptying of the bowel.
16. Take off sleeve. Wash around the stoma or shower to remove fecal drainage. Put on a new pouch.
17. Clean irrigation sleeve, cone tip, and irrigation bag with soap and water. Allow to air dry, and store them in the patient's bathroom.
18. Instruct patient/caregiver in procedure. See Patient Education Guidelines for Colostomy Irrigation.
19. Clean and replace the equipment. Discard disposable items in a trash bag.
20. Document procedure, amount and type of irrigation solution, results of irrigation, condition of stoma and skin, patient instructions and ability to follow through on patient visit report.

Note: https://youtu.be/cwaz-v4-N4g https://youtu.be/m6UMElGvd0U

Follow up activities

1.A patient, who has a colostomy, asks what type of foods they should avoid to decrease odorous gas. You would tell the patient to avoid:

A. Onions, alcoholic beverages, eggs, and cabbage

B. Beef, fried foods, lettuce, and rice

C. Apple, pears, nuts, and wheat

D. Potatoes, peas, carrots, and chicken

2. A patient is 2 days post-opt from an ileostomy placement. Which finding requires immediate nursing action?

A. The stoma is excreting liquid stool.

B. The patient's potassium level is 2.0

C. The stoma is bright red and moist.

D. The patient reports mucoid drainage from the anus.

3. Which type of colostomy can allow a patient to have bowel continence?

A. Descending or Sigmoid Colostomy

B. Ascending or Transverse Colostomy

C. Transverse or Descending Colostomy

D. Ascending or Descending Colostomy

4. The nurse caring for a 2-day postoperative colostomy patient should report immediately if a stoma is assessed as:

A. beefy and red.
B. having swelling.
C. having a small amount of bleeding around it.
D. blue-tinged.

5. The patient says, “I hate this yucky paste under my appliance. I think I will just tape it on.” The nurse’s most informative response to this remark would be which of the following?

A. Taping will not work!
B. Taping will not seal the wafer tight enough to prevent leakage or fill increases.
C. Taping with waterproof tape is just as effective as the paste.
D. Taping is far more irritating to the skin than the paste would be.

6. The patient comes to the industrial nurse and is frantic because the stoma to the colostomy has prolapsed after 1 year post surgery. The nurse’s best counsel would be which of the following?

A. If there are still feces coming from the stoma, it is not blocked. Contact your surgeon for an evaluation.

B. You must come in immediately, because the stoma may completely retract into your abdomen.

C. This is an emergency situation, because it has stenosed.

D. Don’t worry about that. Coughing or sneezing might have caused the prolapse. It will come back out in a few hours.

7. What part of the bowel would a colostomy be placed?

A. Ileum

B. Colon

C. Jejunum

8.When cleaning the skin around the stoma you should use what kind of soap?

A. Dove

B. Ivory

C. Baby Wash

D. Antibacterial

9. True or False:

A. Patients with an ileostomy are at greater risk for dehydration and an electrolyte imbalance.

B. An ileostomy is a surgical opening created to bring the large intestine to the surface of the abdomen

Module

Module Unit: - Gastro Intestinal Tract

Module Title: - liver dialysis

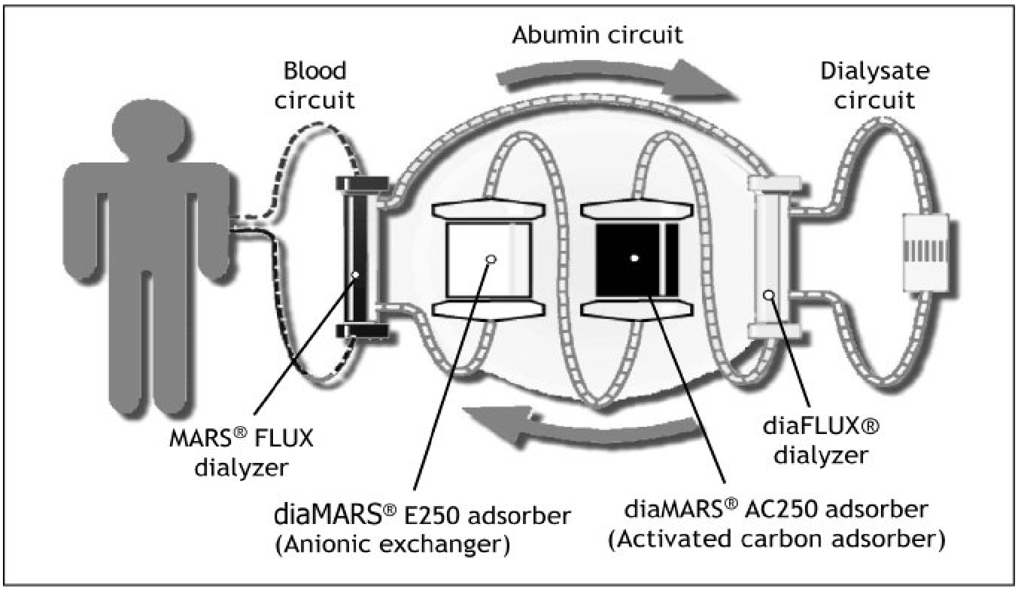
Liver dialysis : is a temporary, artificial, extracorporeal detoxification treatment for severe liver failure that cleans the blood of toxic, protein-bound substances (such as bile acids and ammonia) when the liver can no longer do so. It acts as a "bridge" to support the patient, allowing the liver to recover until a transplant can occur. It operates as a continuous therapy in the ICU for 6–8 hours, using a specialized machine to filter blood through an albumin-rich dialysate, which is then regenerated, allowing the body to recover while awaiting a transplant.

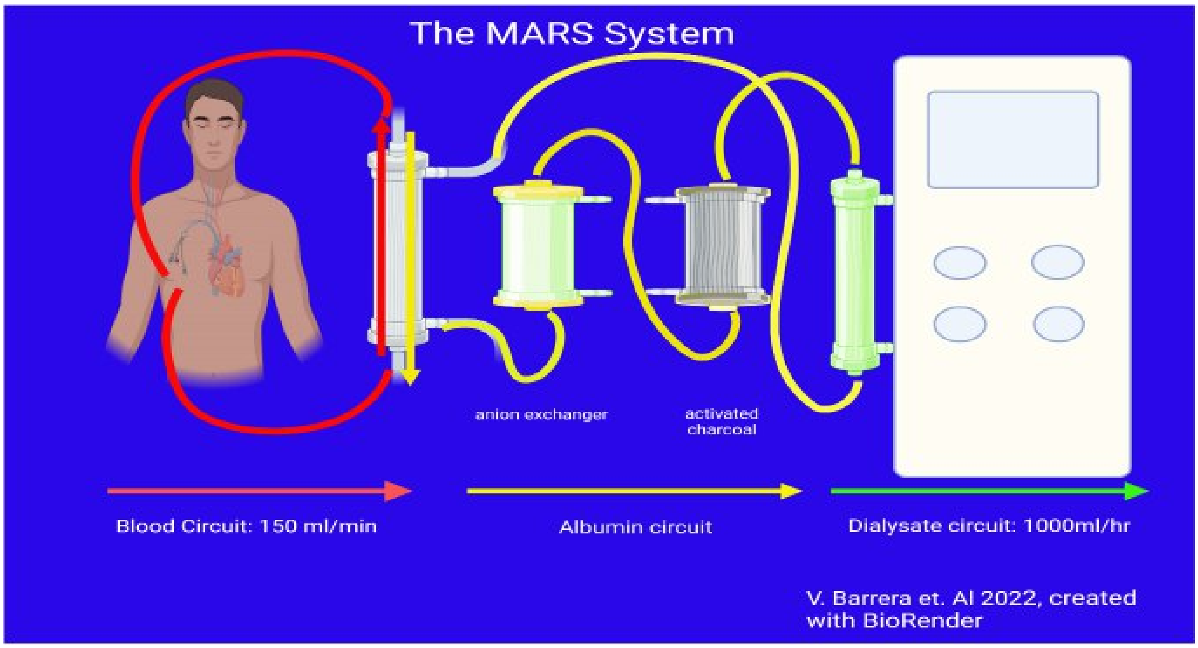
Unlike kidney dialysis which removes water and small solutes, Liver Dialysis must remove protein-bound toxins (like bilirubin and bile acids) and ammonia.

- Purpose:
- To support the liver by "cleaning" the blood to improve neurological status (Hepatic Encephalopathy) and allow the liver time to regenerate or bridge the patient to a transplant.
- Types of Liver Dialysis

1- The Molecular Adsorbent Recirculating System (MARS): the best extracorporeal liver dialysis system in the medical world. Two separate dialysis circuits are used in the MARS blood purification system. MARS is a widely used liver dialysis technique but rarely used in developing countries due to its complicated technique and high cost.

2- Single Pass Albumin Dialysis (SPAD): a simple albumin dialysis technique as it uses standard renal replacement therapy machines without the interruption of additional perfusion pump systems like MARS.





Nursing Action	Rationale
Pre-Procedure Assessment	
Neurological Baseline: use the Glasgow Coma Scale (GCS).	Liver failure causes ammonia buildup, leading to confusion or coma
Coagulation Profile: check INR, PT, and platelets.	Patients with liver failure have a high risk of bleeding

Vascular Access: Ensure a large-bore double-lumen catheter (usually internal jugular) is patent.	Observe for any inflammation signs as swelling, tenderness.
Prepare and check the main circuits (blood or patient, albumin, and dialysis) and explain the procedure to the patient if conscious or to his family	
During session	
Continuous Monitoring for ● Blood pressure (hypotension is common) ● Heart rate ● Oxygen saturation ● Level of consciousness Observe for complications as bleeding: ● Catheter site, gums , nose ● Bruising Machine-related problems: ● Air bubbles ● Clotting ● Alarms Monitor anticoagulation therapy and document all medications given during dialysis.	
Post- dialysis nursing care	
Reassess: ● Vital signs (specially, blood pressure and temperature) ● Mental status improvement Inspecting catheter site Follow up post-dialysis lab results Documentation: ● Duration of session	Liver dialysis can cause heat loss, leading to shivering, which increases metabolic demand on an already stressed liver.

Module Title: - liver biopsy

Time Allocated: - one-hour theory / 30 Minute clinical practice.

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with liver biopsy.
- Implements standardized protocol and guidelines when providing nursing care for patients with liver biopsy.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with liver biopsy.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with liver biopsy.
- Develop a teaching plan for patient with liver biopsy.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with liver biopsy.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Outlines:

1. Definition.
2. Indications.
3. Contraindications.
4. Types.
5. Equipment.
6. Procedure.
7. Complications.

Liver Biopsy

Definition:

Liver biopsy is a diagnostic test, which involves the removal of a small quantity of tissue from the liver, usually done under local anesthetic, to allow microscopic examination of the liver.

Indications:

1. Evaluation of abnormal liver biochemical tests and hepatomegaly.
2. Evaluation and staging of chronic hepatitis.
3. Evaluation the type and extent of drug induced liver injury.
4. Identifying and determination of the nature of intrahepatic masses.
5. Diagnosis of multisystem infiltrative disease.
6. Evaluation the effectiveness of the therapies for liver disease.
7. Evaluation of liver test abnormalities following transplantation.

Contraindications

- Absolute

- History of un explained bleeding
- Prothrombin time < 3-4s over control
- Platelets > 60.000/mm^3.
- Prolonged bleeding time ≥ 10 min.
- Un availability of blood transfusion support.

- Suspected hemangioma.
- Uncooperative patients.

- <u>Relative</u>
    - Ascites.
    - Infections in right pleural cavity.
    - Infection below right diaphragm.
    - Morbid obesity.

<u>Types of liver biopsy:</u>

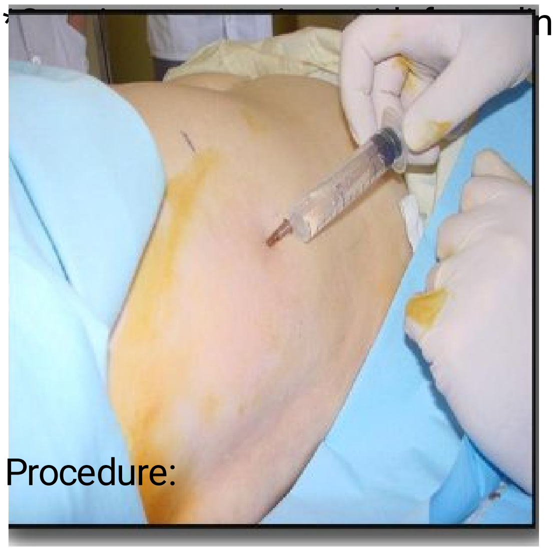
- Percutaneous liver biopsy
    - Ultrasound guided liver biopsy.
    - Percussion guided liver biopsy.
- Rare
    - Trans jugular liver biopsy.
    - Tran's femoral liver biopsy.
    - Laparoscopic liver biopsy

Site of liver biopsy: the biopsy site is in the seventh or eighth intercostal space in the mid axillary line.

<u>Equipment:</u>

*Sterile gloves *Dressing kit *Antiseptic solution

*Local anesthetic *Biopsy needle *Needle and syringe



Procedure:

Nursing Action	Rationale
Preparatory phase:	
Ascertain the results of coagulation tests (prothrombin time, partial thromboplastin time, and platelet count) are available and that a compatible donor blood is available.	Many patients with liver diseases have clotting defects and are at risk for bleeding
Check for signed consent; confirm that informed consent has been provided.	Ensure that the patient consent to this invasive procedure
Measure and record the patient's pulse, respirations, and blood pressure immediately before biopsy.	Prebiopsy values provide a basis on which to compare the patient vital signs and evaluate status after procedure
Describe to the patient in advance: steps of the procedure; sensations expected; after-effects anticipated; restrictions of activity and monitoring procedures to follow.	Explanations allay fears and ensure cooperation.
	Encouragement and support of the nurse enhance comfort and promote a sense of security.
Performance phase:	
5- Support the patient during the procedure.	The skin at the site of penetration will be cleansed and a local anesthetic will be infiltrated.
Expose the right side of the patient's upper abdomen (right hypochondria)	Holding the breath immobilizes the chest wall and the diaphragm; penetration of the diaphragm thereby is
Instruct the patient to inhale and exhale deeply several times, finally to exhale, and	

avoided, and

hold breath at the end of expiration.

The physician promptly introduces the biopsy needle by way of the intrathoracic (intercostal) or transabdominal (subcostal) route, penetrates the liver, aspirates, and withdraws.

the risk of lacerating the liver is minimized.

The patient often continues holding his or her breath because of anxiety.

8- Instruct the patient to resume breathing.

Follow up phase:

9- Immediately after the biopsy, assist the patient to be on the right side; place a pillow under the costal margin, and caution the patient to remain in this position, and immobile for several hours. Instruct the patient to avoid coughing or straining.

In this position, the liver capsule at the sight of penetration is compressed against the chest wall, and the escape of blood or bile through the perforation is prevented.

10- Measure and record the patient's pulse, respiratory rate, and blood pressure at 10- to 15-minutes intervals for the first hour, then every 30 minutes for the next 1 to 2 hours or until the patient's condition stabilizes.

Changes in vital signs may indicate bleeding, sever hemorrhage, or bile peritonitis, the most frequent complications of liver biopsy.

11- If the patient is discharged after the procedure, instruct the patient to avoid heavy lifting and strenuous activity for 1 week

Activity restrictions reduce the risk of bleeding at the biopsy puncture site.

Complications:

- Bile Peritonitis
- Severe bleeding
- Accidental injury of the surrounding organ
- Sepsis
- Pain

Note: https://youtu.be/tKjxMw81L2k

Follow up activities

1.Define liver biopsy:

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2.Mention nursing care needed before performing liver biopsy:

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3.List indication of liver biopsy:

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4. The appropriate site for inserting liver biopsy needle is

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5. The patient lies on the right site after liver biopsy to

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Module

Module Unit: - Gastro Intestinal Tract

Module Title: - Paracentesis

Time Allocated: - one-hour theory / 30 Minute clinical practice

(Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with paracentesis.
- Implements standardized protocol and guidelines when providing nursing care for patients with paracentesis.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with paracentesis.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with paracentesis.
- Develop a teaching plan for patient with paracentesis.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with paracentesis.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Outlines:

1.Definition.

2.Indications.

3.Contraindications.

4.Equipment.

5.Procedure.

6.Complications.

Paracentesis

Definition: is a procedure performed to obtain a small sample of or drain ascitic fluid for either diagnostic or therapeutic purposes.

Ascitic fluid may be used to help determine the etiology of ascites. The means for characterizing ascites is the serum-ascitic albumin gradient (SAAG)

SAAG = the albumin concentration of a serum specimen - the albumin concentration of the ascitic fluid.

The SAAG correlates directly with portal pressure: if

The SAAG is >1.1 g /dl indicate that ascites is related to portal hypertension.

The SAAG is < 1.1 g /dl indicate that ascites is related to other causes.

Indications:

Is used for a number of reasons:

- To relieve abdominal pressure from ascites.
- To diagnose spontaneous bacterial peritonitis and other infections (e.g. abdominal TB).
- To diagnose metastatic cancer.
- To diagnose blood in peritoneal space in trauma.

Contraindications:

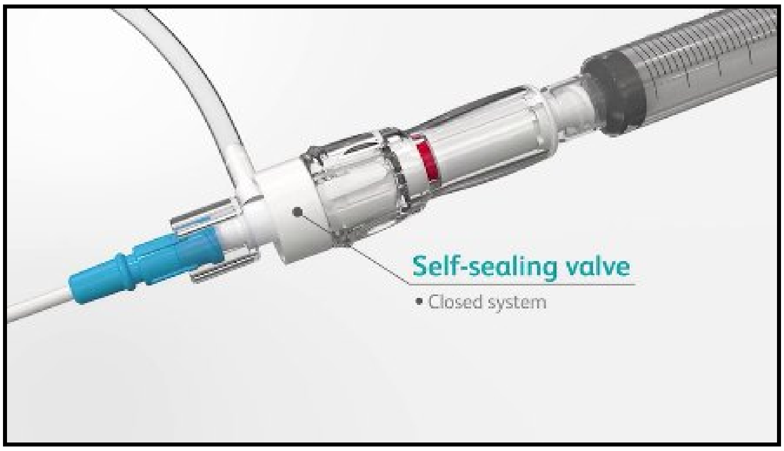
- Absolute contraindication:
    - Disseminated intravascular coagulopathy and sever thrombocytopenia

As the risk of bleeding may be increased if:

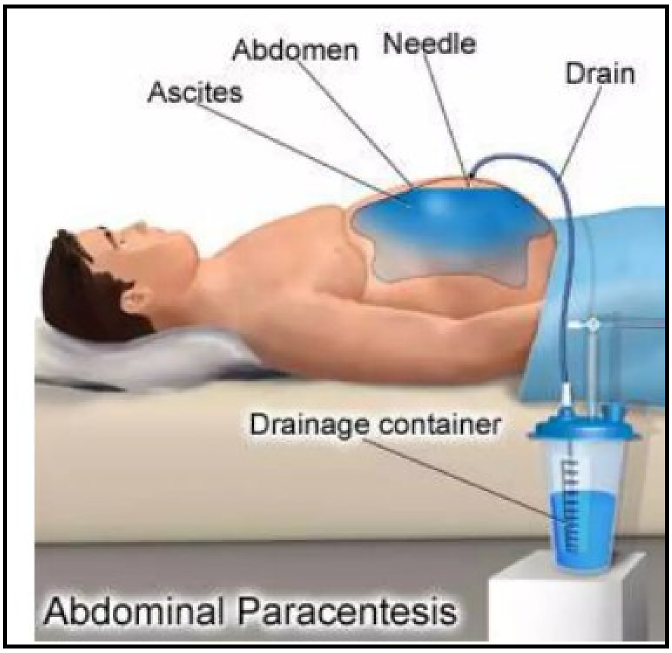
- Prothrombin time > 21 seconds
- International normalized ratio > 1.6
- Platelet count < 50,000 per cubic millimeter.
  - Acute abdomen that requires surgery.
- Relative contraindications are:
  - Pregnancy.
  - Distended urinary bladder.
  - Abdominal wall cellulitis.
  - Distended bowel.
  - Intra-abdominal adhesions.

Equipment:

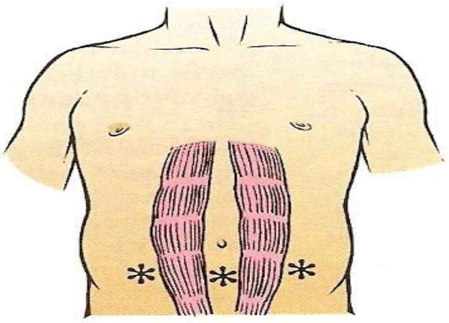
- Antiseptic swab sticks.
- Drape.
- Lidocaine 1%.
- Syringe 10 ml, 60 ml.
- Injection needles 22-25 gauges.
- Eight- French catheter with 3-way stopcock, self-sealing valve, and a 5-ml luer-lock syringe.

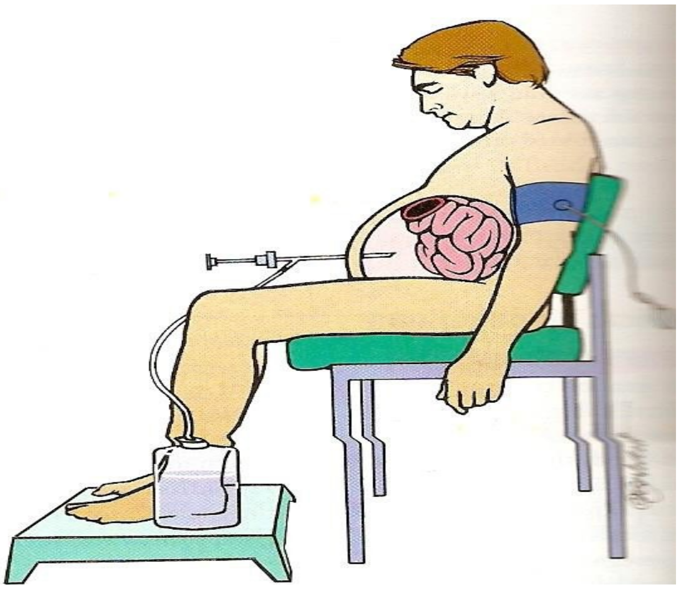


- Introducer needle 20 gauge.
- Adhesive dressing.
- Tubing set with roller clamp.
- Collection bottle vacuum.
- Specimen bottles and laboratory forms



- Gauze, 4x4 inch.





Procedure:

Nursing Action	Rationale
Preparatory phase:	
1-check the patient's coagulation profile.	1- This may reduce the patient's fear and anxiety.
2. Explain procedure to patient.	
3- Record patient vital signs.	2- Provides baseline values for later comparison.
4- Instruct the patient to void.	3- This will lessen the danger of accidentally penetrating the bladder with the needle or trocar.
5- Place patient in upright position on edge of bed with feet supported on stool, or place in chair. Fowler's position should be used for the patient confined to bed.	4- The patient is more comfortable, and a steady position can be maintained.
6- Drape the patient with sheet exposing abdomen.	5- Minimize exposure of patient and keeps patient warm.

Procedure: (continue)

Nursing Action	Rationale
Performance phase:	
1- Assist in preparing skin with antiseptic solution.	1- This is considered a minor surgical procedure that requires aseptic precautions.
2- Open sterile tray and package of sterile gloves; provide anesthetic solution.	
3- Have collection bottle and tubing available.	
4- Assess pulse and respiratory statue frequently during the procedure; watch for pallor, cyanosis, or syncope.	4- Indicates shock. Keep emergency drugs available.
5- Physician administers local anesthesia and introduces needle or trocar.	
1- Needle or trocar is connected to tubing and vacuum bottle or syringe; fluid is slowly drained from peritoneal cavity.	
2- Apply dressing when needle is withdrawn.	2- Documentation is important for continuity of care.
Follow-up phase:	
1- Assist the patient to a comfortable position after treatment.	
2- Record amount and characteristics of fluid removed, number of	

condition during treatment.

3- Check blood pressure and vital signs every 1/2 hour for 2 hours, every hour for 4 hours, and every 4 hours for 24 hours.	·Close observation will detect poor circulatory adjustment and possible development of shock.
*4- Usually, a dressing is sufficient; however, if the trocar wound appears large, the physician may close the incision with sutures.	·To prevent blood loss and aid healing.
5- Watch for leakage or scrotal edema (in men) after paracentesis.	·If seen, report at once.

Complications

- Persistent leakage of ascitic fluid at the needle insertion site. This can often be addressed with a single skin suture.
- Abdominal wall hematoma or other bleeding.
- Infection.
- Perforation of surrounding vessels or viscera (extremely rare).
- Hypotension after large volume fluid removal (more than 5 L to 6 L). Albumin is often administered after removal of more than 5 L of fluid to prevent this complication.
- Dilutional hyponatremia - Catheter laceration and loss in abdominal cavity
- Laceration of major blood vessel (aorta, mesenteric artery, iliac artery).
- Hepatorenal syndrome.

Note: https://youtu.be/O8wOhjFOW5A

Follow up of

activities Choose the correct answer:

1. which of the following reflects abdominal paracentesis:

Is a non-invasive procedure.

b) Can be performed under ultrasound guidance.
c) Is used only for diagnostic purposes.
d) Can be performed without exercising the aseptic technique.
e) Is performed by the nurse.

2. What is ascites meaning?

a) Is most commonly caused by malignancy?
b) Is defined as presence of excess fluid in the pleural cavity.
c) Can be 'transudate' or 'exudate'.
d) Can only be detected by ultrasound scan of the abdomen.
e) Occur only in patients with liver cirrhosis

3. Which of these describes the classical clinical manifestation of large-volume ascites?

a) Shifting dullness is not a feature.
b) It can only be detected by CT scan of the abdomen
c) It is always 'exudate'.
d) It is associated with a positive fluid thrill.
e) Its amount is about 3 liters.

4. Which of the following is done when performing an abdominal paracentesis

a. It is unnecessary to obtain informed consent from the patient
b. It is crucial to exercise the aseptic technique
c. It can be done by final year medical students when other medical staff

are not available

d. Monitoring of vital signs following the procedure should only be carried Out upon request by doctors

e. The patient is lying in supine position

Module

*Module Unite: Endoscopic procedures used in gastrointestinal tract

*Module Title: Endoscopic procedures used in gastrointestinal tract

*Time Allotted:1hour theory and 30 Minute clinical practice (Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with liver biopsy.
- Implements standardized protocol and guidelines when providing nursing care for patients with liver biopsy.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with liver biopsy.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with liver biopsy.
- Develop a teaching plan for patient with liver biopsy.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with liver biopsy.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Outlines:

- Definition of endoscopy
- Types of endoscope
- Uses of endoscope
- The nursing care of endoscope

Endoscopic procedures used in gastrointestinal tract

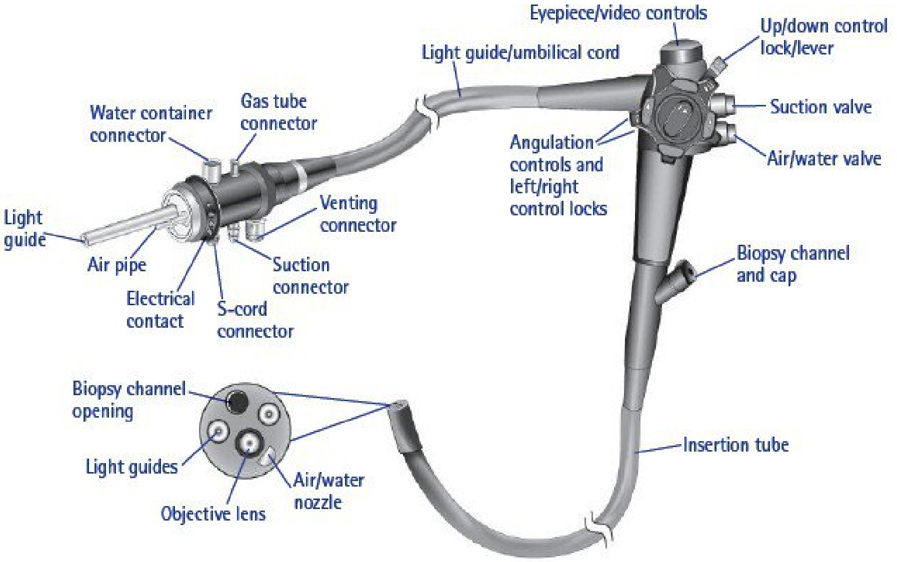
Endoscopy

Endoscopic investigations allow direct visualization of the mucosal lining and organs of the gastrointestinal tract using rigid or flexible scopes equipped with a light source and camera

Endoscopic Procedures

Endoscopic procedures used in gastrointestinal tract assessment include:

1. Upper Gastrointestinal Fibroscopy/ Esophagogastroduodenoscopy (EGD)
2. Fiberoptic Colonoscopy
3. Anoscopy, Proctoscopy, and Sigmoidoscopy



▲ Flexible endoscopes have glass fiber bundles in the tubing that transmit a picture back to a camera or eyepiece. Flex endoscopes also have a distal end that moves from two to four different directions for better viewing of the body ca (Courtesy of Endoscopy Support Services, Inc, www.endoscopy.com.)

Upper Gastrointestinal Fibroscopy/ Esophagogastroduodenoscopy

Fibroscopy of the upper GI tract allows direct visualization of the esophageal, gastric, and duodenal mucosa through a lighted endoscope (gastroscope)

EGD can be used in:

Diagnostic Purposes:

- when esophageal, gastric, or duodenal disorders or inflammatory, neoplastic, or infectious processes are suspected.
- to evaluate esophageal and gastric motility
- to collect secretions and tissue specimens for further analysis.
- To obtain images through the scope to document findings.
- Electronic video endoscopes also are available that attach directly to a video processor, converting the electronic signals into pictures on a television screen. This allows larger and continuous viewing capabilities, as well as the simultaneous recording of the procedure.

Therapeutic Purposes

- Upper GI fibroscopy also can be a therapeutic procedure when combined with other procedures. Therapeutic endoscopy can be used to remove common bile duct stones, dilate strictures, and treat gastric bleeding and esophageal varices. Laser-compatible scopes can be used to provide laser therapy for upper GI neoplasms. Sclerosing solutions can be injected through the scope in an attempt to control upper GI bleeding

Nursing Interventions

- The patient should be NPO for 8 hours prior to the examination.
- Before the introduction of the endoscope, the patient is given a local anesthetic gargle or spray.
- Midazolam (Versed), a sedative that provides moderate sedation and relieves anxiety during the procedure, may be administered.
- Atropine may be administered to reduce secretions, and glucagon may be administered to relax smooth muscle.
- The patient is positioned in the left lateral position to facilitate clearance of pulmonary secretions and provide smooth entry of the scope.

After gastroscopy

- Assessment includes: level of consciousness, vital signs, oxygen saturation, pain level, and monitoring for signs of perforation (ie, pain, bleeding, unusual difficulty swallowing, and rapidly elevated temperature).
- After the patient's gag reflex has returned, lozenges, saline gargle, and oral analgesic agents may be offered to relieve minor throat discomfort.

- Patients who were sedated for the procedure must remain in bed until fully alert.
  - After moderate sedation, the patient must be transported home with a family member or friend if the procedure was performed on an outpatient basis.
  - Someone should stay with the patient until the morning after the procedure.
  - Because of sedation, many patients will not remember post procedure instructions. For this reason, discharge and follow-up instructions are provided to the person accompanying the patient home, as well as to the patient.

<u>Fiberoptic Colonoscopy</u>

Direct visual inspection of the large intestine (anus, rectum, sigmoid, transcending and ascending colon) is possible by means of a flexible fiberoptic colonoscope.

<u>Fiberoptic colonoscopy can be used in:</u>

Diagnostic Purposes:

- This procedure is used commonly as a diagnostic aid and screening device. It is most frequently used for cancer screening and for surveillance in patients with previous colon cancer or polyps. In addition, tissue biopsies can be obtained as needed, and polyps can be removed and evaluated.
- Other uses of colonoscopy include the evaluation of patients with diarrhea of unknown cause, occult bleeding, or anemia; further study of abnormalities detected on barium enema; and diagnosis, clarification, and determination of the extent of inflammatory

Therapeutic Purposes

- Therapeutically, the procedure can be used to remove all visible polyps with a special snare and cautery through the colonoscope.
- Many colon cancers begin with adenomatous polyps of the colon; therefore, one goal of colonoscopic polypectomy is early detection and prevention of colorectal cancer.
- This procedure also can be used to treat areas of bleeding or stricture. Use of bipolar and unipolar coagulators, use of heater probes, and injections of sclerosing agents or vasoconstrictors are all possible during this procedure.
- Laser-compatible scopes provide laser therapy for bleeding lesions or colonic neoplasms.
- Bowel decompression (removal of intestinal contents to prevent gas and fluid from distending the coils of the intestine) can also be completed during the procedure.

Nursing Interventions:

The success of the procedure depends on how well the colon is prepared and on adequate sedation. Adequate colon cleansing provides optimal visualization and decreases the time needed for the procedure.

<u>Before the procedure</u>

1. Cleansing of the colon can be accomplished in various ways:

- The physician may prescribe a laxative for 2 nights before the examination and a Fleet's or saline enema until the return is clear the morning of the test.
- However, more commonly, polyethylene glycol electrolyte lavage solutions are used as intestinal lavages for effective

cleansing of the bowel.

- The patient maintains a clear liquid diet starting at noon the day before the procedure. Then the patient ingests the lavage solution orally at intervals over 3 to 4 hours. If necessary, the nurse can give the solution through a feeding tube if the patient cannot swallow.
- The use of lavage solutions is contraindicated in patients with intestinal obstruction or inflammatory bowel disease.
- the use of lavage solutions, bowel cleansing is fast (rectal effluent is clear in about 4 hours) and is tolerated fairly well by most patients.
- A sodium phosphate tablet (Osmoprep, Visicol) can be used for colon cleansing prior to colonoscopy. Dosing consists of 32 tablets: 20 tablets (4 tablets every 15 minutes with 8 ounces of any clear liquid (water, any clear carbonated beverage, or juice) on the evening prior to the examination, and 12 tablets (taken in the same manner) on the morning of the examination.
- Side effects of the electrolyte solutions include nausea, bloating, cramps or abdominal fullness, fluid and electrolyte imbalance, and hypothermia (patients are often told to drink the preparation as cold as possible to make it more palatable).
- the nurse advises the patient with diabetes to consult with his or her physician about medication adjustment to prevent hyperglycemia or hypoglycemia resulting from the dietary modifications required in preparing for the test.
- The nurse also instructs all patients, especially the elderly, to maintain adequate fluid, electrolyte, and caloric intake while undergoing bowel cleansing.

- Special precautions must be taken for some patients. Implantable defibrillators and pacemakers are at high risk of malfunction if electrosurgical procedures (ie, polypectomy) are performed in conjunction with colonoscopy. A cardiologist should be consulted before the test is performed, and the defibrillator should be turned off. These patients require careful cardiac monitoring during the procedure.
- Colonoscopy cannot be performed if there is a suspected or documented colon perforation, acute severe diverticulitis, or fulminant colitis.

2. Patients with prosthetic heart valves or a history of endocarditis require prophylactic antibiotics before the procedure.
3. Informed consent is obtained by the practitioner before the patient is sedated. Before the examination.
4. An opioid analgesic or sedative agent (eg, midazolam [versed]) is administered to provide moderate sedation and relieve anxiety during the procedure.
5. Glucagon may be administered, if needed, to relax the colonic musculature and to reduce spasm during the test.
6. Elderly or debilitated patients may require a reduced dosage of the analgesic or sedative agent to decrease the risks of over sedation and cardiopulmonary complications.

<u>During the procedure</u>

- Colonoscopy is performed while the patient is lying on the left side with the legs drawn up toward the chest. The patient's position may be changed during the test to facilitate advancement of the scope

- the patient is monitored for changes in oxygen saturation, vital signs, color and temperature of the skin, level of consciousness, abdominal distention, vagal response, and pain intensity.

After the procedure

- patients are maintained on bed rest until fully alert.
  - Some patients have abdominal cramps caused by increased peristalsis stimulated by the air insufflated into the bowel during the procedure.
  - Immediately after the test, the patient is monitored for signs and symptoms of bowel perforation (eg, rectal bleeding, abdominal pain or distention, fever, focal peritoneal signs).
  - Because of the amnesic effects of midazolam, it is important to provide written instructions, because the patient may be unable to recall verbal information.
  - If the procedure is performed on an outpatient basis, someone must transport the patient home.
  - After a therapeutic procedure, the nurse instructs the patient to report any bleeding to the physician.

Complications:

Complications during and after the procedure can include

1. Vasovagal reactions
2. Cardiac dysrhythmias
3. Respiratory depression resulting from the medications administered;

4. Circulatory overload or hypotension resulting from over hydration or under hydration during bowel preparation.

Anoscopy, Proctoscopy, and Sigmoidoscopy

Endoscopic examination of the anus, rectum, and sigmoid and descending colon is used to evaluate chronic diarrhea, fecal incontinence, ischemic colitis, and lower GI hemorrhage and to observe for ulceration, fissures, abscesses, tumors, polyps, or other pathologic processes.

Nursing Interventions:

Before the procedure:

- These examinations require only limited bowel preparation, including a warm tap water or Fleet's enema until returns are clear. Dietary restrictions usually are not necessary, and sedation usually is not required.

During the procedure:

- The patient assumes a comfortable position on the left side with the right leg bent and placed anteriorly
- The nurse monitors vital signs, skin color and temperature, pain tolerance, and vagal response.

After the procedure:

- The nurse monitors the patient for rectal bleeding and signs of intestinal perforation (ie, fever, rectal drainage, abdominal distention, and pain).
- On completion of the examination, the patient can resume his or her regular activities and diet.

Follow Up Activities

Complete the following:

1. What are the endoscopic procedures used in gastrointestinal tract assessment?

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2. What are the uses of Esophagogastroduodenoscopy (EGD)?

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3. Give notes about nursing interventions in Anoscopy, Proctoscopy, and Sigmoidoscopy:

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Module

*Module Unite: Immobilization

*Module Title: Cast Care

*Duration: 1hour theory and.30 Minute clinical practice (Demonstration and re- demonstration)

Objectives:

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with cast.
- Implements standardized protocol and guidelines when providing nursing care for patients with cast.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with cast.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with cast.
- Develop a teaching plan for patient with cast.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with cast.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Cast care

Outlines:

- Definition of cast
- Purposes of cast
- Types of cast
- Types of cast materials
- Principles of cast application
- The equipment that used in cast application and cast removal
- Technique of cast application
- Technique of cast removal

Definition of cast

Cast is a rigid external an immobilizing device made up of layers of bandage molded to the contours of the affected body part.

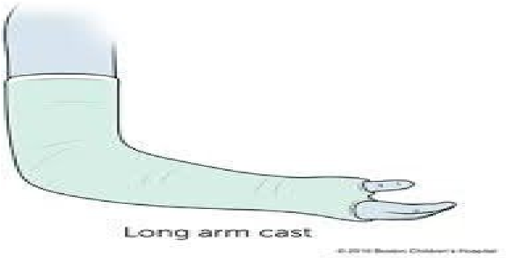
Purposes of cast:

- To immobilize and hold bone fragments in the reduction.
- To apply uniform compression of soft tissue.
- To permit early mobilization.
- To correct and prevent deformities.
- To support and stabilize weak joints.

Types of cast:

Short arm cast: extends from below the elbow to the palm, secured around the base of the thumb. If the thumb is included, it is known as a thumb Spica or gauntlet cast. Used for fracture of hand and wrist.

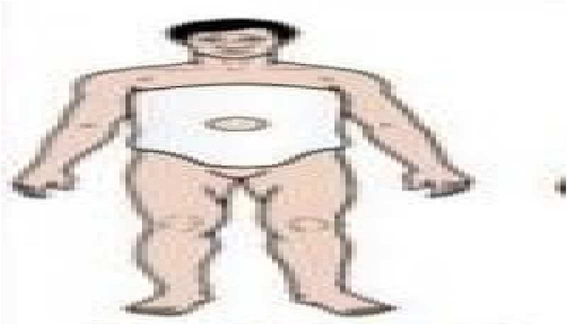
Long arm cast: extends from the upper level of the axillary fold to the palm. The elbow usually immobilized at right angle. Used for fracture of forearm, elbow or humerus.



Short leg cast: extends from below the knee to the base of the toes. The foot is flexed at a right angle in a neutral position. Used for fracture of foot, ankle or distal tibia or fibula.

Long leg cast: extends from the junction of the upper and middle third of the thigh to the base of the toes. The knee may be slightly flexed. Used for fracture of distal femur, knee or knee dislocation.

Body cast: Encircles the trunk. Used to stable spine injuries or the thoracic or lumbar spine



Shoulder Spica cast: A body jacket that encloses the trunk and the shoulder and elbow.

Shoulder spica cast

Hip Spica cast: Encloses the trunk and a lower extremity. A double hip Spica cast includes both legs. Used for fracture of femur or pelvis.

Types of cast material

1. Non plaster cast

Generally referred to as fiberglass cast, it is a synthetic material used for casts that is lighter and has shorter drying time than plaster of Paris.

Non plaster cast

Plaster of Paris

Advantages of non-plaster cast

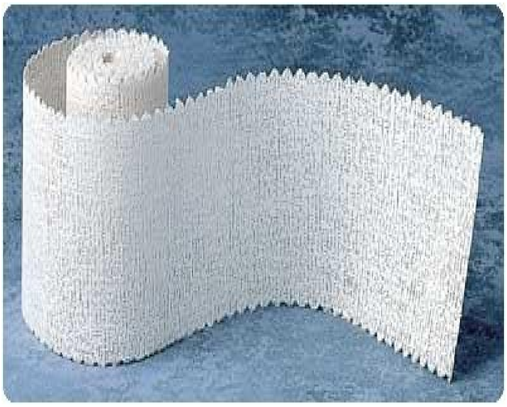
- They are lighter in weight, stronger, water resistant, and durable.
- Non plaster casts are porous and therefore diminish skin problems.
- When wet, they are dried with a hair drier on a cool setting
- Drying time is 10 to 15 minutes and the cast can withstand weight bearing 30 minutes after application.

2. Plaster cast

The traditional cast is made of plaster. Rolls of plaster bandage are wet in cool water and applied smoothly to the body. A crystallizing reaction occurs, and heat is given off.

Initially the wet cast is hot, and the heat may cause edema of the underlying tissues from the increased circulation. The cast quickly become damp and cold.

It dries after about 24 to 72 hours, depending on the size and location.



Advantages of plaster cast

- Moulds very smoothly to body contour
- It is the least expensive type of cast to be used.

<u>Cast care:(health teaching for patient)</u>

- Keep plaster cast dry.
- Do not remove any padding.
- Do not insert any foreign object inside the cast.
- Do not bear weight on a new plaster cast for 48 hours (with fibroglass cast, may be less than one hour).
- Do not cover the cast with plastic for prolonged period.
- Report the health care provider if there is swelling, discoloration of toes or fingers, pain during motion, and burning or tingling under the cast.
- Elevate affected extremity on pillows (danger of edema is usually 24 to 48 hours).

*Equipment that used in cast application

•Appropriate size and quantity of cast materials

•Appropriate size and quantity of cast padding or stockinette

- Pail for water
- Gloves and aprons
- Plastic drapes or sheet
- Scissors
- Cart, chair, fracture table

Technique for cast application:

Steps	Rational
Wash hands	Reduces transmission of infection
Wear gloves and apron	Plaster is abrasive to the skin
Support extremity or body part to be casted.	Minimizes movement; maintains reduction and alignment; increases comfort.
Position and maintain part to be casted in normal anatomical position that indicated by physician during casting procedure.	Facilitates casting; reduces incidence of complications (e.g., malunion, nonunion, contracture).
Drape patient.	Avoids undue exposure; protects other body parts from contact with casting materials.
Wash and dry part to be casted.	Reduces skin breakdown.
Place knitted material (e.g., stockinet) over part to be casted. • Apply in smooth and non-constrictive manner. • Allow additional material	Protects skin from casting materials. Protects skin from pressure.

Wrap soft, nonwoven roll padding smoothly and evenly around part.	Protects skin from pressure of cast.
• Use additional padding around bony	Protects skin at bony prominences.

prominences:	Protects superficial nerves.
Apply plaster or non-plaster casting material evenly on body part.	Creates smooth, solid and well-contoured cast.
• Choose appropriate width bandage.	
• Overlap preceding turn by half the width of the bandage.	Facilitates smooth application.
• Use continuous motion, maintaining constant contact with body part.	Creates smooth, solid, immobilizing cast.
• Use additional casting material (splints) at joints and at points of anticipated cast stress.	Shapes cast properly for adequate support and strengthens cast.
“Finish” cast:	
• Smooth edges.	Protects skin from abrasion.
• Trim and reshape with cast cutter.	Assures full range of motion of adjacent joints.
Remove particles of casting materials from skin.	Prevents particles from loosening and sliding underneath cast.
Support cast during hardening.	
• Handle hardening casts with palms of hands.	Avoids denting of cast and development of pressure areas.
• Support cast on firm smooth surface.	
• Don’t rest cast on hard surfaces or on sharp edges	
• Avoid pressure on cast.	
Promote drying of cast.	
• Leave cast uncovered.	Facilitates drying.

assist patient to transfer to bed if necessary	To prevent patient injury
Clean equipment that used, return to usual places and discard used materials	To prevent transmission of infection
Wash hands	
Document in patient record	

Cast removal

Equipment that used in cast removal

•

One cast cutter (cast saw)

•One cast spreader

•Scissors and Skin lotion

•Basin, water, washcloths, towels

•Appropriate supportive device such as elastic stocking.

*Technique of cast removal

Steps	Rational
Wash hands	Reduces transmission of infection
Wear gloves and apron	Plaster is abrasive to the skin

Inform the patient about the procedure.	Facilitates cooperation and reduces fear about the procedure.
Reassure patient that the electric saw or cast cutter will not cut skin	Reduces anxiety. (Explains that blade oscillates to cut cast and vibrations will be felt.)
Wear eye protection (patient and operator of the cast cutter).	Protects eyes from flying cast particles.
Bivalve cast using a series of alternating pressures and linear movements of blade along the line to be cut.	Cuts cast in halves. Avoids burning sensation from prolonged contact of oscillating blade with padding.
Cut padding with scissors.	Releases all of the casting materials.
Support body part where cast is removed	Reduces stress on body part

Gently wash and dry area that has been immobilized. Apply lotion as prescribed	Removes dead skin that has accumulated during immobilization.
Teach patient to avoid rubbing and scratching the skin	Prevents skin breakdown.
Assist in the application of supportive devices as prescribed	For support of weak muscles and unstable joints in the affected part.
Collaborates with physical therapist to teach patient to resume active use of body part gradually	Reduce stiffness, restore muscle strength and function
Teach patient to elevate the extremity or using elastic bandage if prescribed.	Improve circulation and venous return and control swelling
Clean equipment that used, return to usual places and discard used materials	To prevent transmission of infection
Wash hands and Document in record	

Note: https://youtu.be/PM_fwwVTuXs

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Module

*Module Unite: Immobilization

*Module Title: Traction Care

*Time allotted: 1hour theory and 30 Minute clinical practice (Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with liver biopsy.
- Implements standardized protocol and guidelines when providing nursing care for patients with liver biopsy.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with liver biopsy.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with liver biopsy.
- Develop a teaching plan for patient with liver biopsy.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with liver biopsy.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Traction Care

Outlines:

1) Definition of traction
2) Purposes of traction
3) Principles of traction
4) Types of traction
5) Comparison between skin and skeletal traction
6) Essential guideline data of traction
7) The equipment that are used in application of skin and skeletal traction
8) Technique of application of skin traction
9) Technique of skeletal traction
10) Nursing care for patient with traction

*Definition of traction

Traction is the force applied in specific direction to a part of the body to overcome the natural force.

Purpose of traction:

1. To reduce and/or immobilize a fracture for healing.
2. To regain normal length and alignment of involved bone.
3. To correct, reduce, or prevent skeletal deformities and contracture.
4. To relieve pressure on the nerve, especially spinal.
5. To decrease muscle spasm.

*Principles of effective traction:

1- Counter traction must be used to achieve effective traction. Counter traction is the force acting in the opposite direction.

2- Traction must be continuous to be effective in reducing and immobilizing fracture

3- The skeletal traction is never interrupted unless intermittent traction is prescribed.

4- Any factors that might reduce the effective pull or alter its resultant line of pull must be eliminated

5- The patient must be in good body alignment in the center of the bed when traction is applied.

6- Robes must be unobstructed.

7- Weight must hang free and not rest on the bed or floor.

8- Knots in the robe or the footplate must not touch the pulley or the foot of the bed.

*Types of traction:

1- Skin traction: traction is applied directly on the skin. The most common method of traction and applied to the skin without pins or wires that break the skin's integrity.

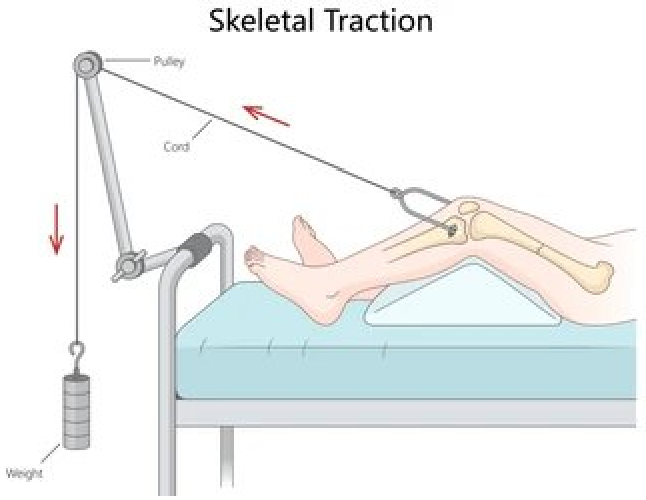
Examples of skin traction is Buck traction that is used for hip and knee contractures and alignment of hip fracture.

Weight used during skin traction should not be more than 5 to 10 Ib (2.5–4.5 kg) to prevent injury to the skin.

2- Skeletal traction: traction is applied directly to the bone by means of metal pin or wire.

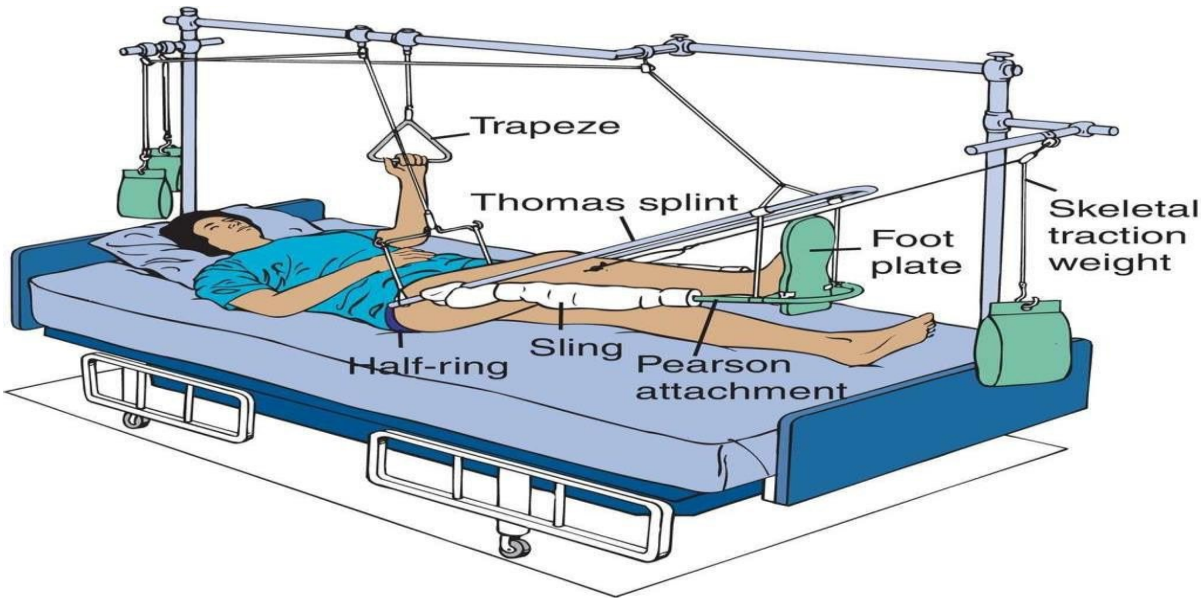
Skeletal traction provides a strong, steady, continuous pull and can be used for prolonged period of time.

Heavier weights can be used with skeletal traction, usually from 15 to 30 Ib (7-14 kg).



a-Straight or running traction: - applying the pulling force in a straight line with the body part resting on the bed.

b- Balanced suspension traction: supports the affected extremities off the bed and allows for some patient movement without disruption of the line of pull.



*Equipment of skin traction:

- Bed frame for attachment of traction or portable frames that attach to bed
- Ropes, pulleys, weights, weight holder
- Spreader bars
- Heel or elbow protectors
- Boots / straps for Buck's traction
- Pelvic strap / belt for pelvic traction
- Elastic bandages and
- Adhesive strips.

*Technique of skin traction:

Steps	Yes	No
1-Wash hands		
2- explain procedure to client		
3-prepare client and area of body to be in traction		
4-Position client as requested by physician		
5- Assist with application of equipment (head halter, adhesive strips and elastic bandages, pelvic belt)		
6- Nurse asked to hold client in desired position or apply halter, strips, elastic bandages while physician and other assistants hold client's tissues in desired positions		
7-Assist with attachment of spreader bars, ropes, pulleys. Ropes are tied securely in knots and passed in grooves of pulleys to weights		

8-When all traction materials and spreader bars are in place, weights are placed on weight holder and hooked to loop in rope and lowered slowly and gently until rope is taught. Physician determines exact amount of weight to be applied
and position to be maintained for majority of time by client
9-Ask client how traction is affected injured tissues if client
is able to respond
10- For safety, raise side rails as appropriate.

11- Gather unused materials and return to storage areas

12- Wash hands

Equipment of skeletal traction:

- Overhead frame for attachment of traction
- Ropes, pulleys, weights and weight holder
- Pearson attachment
- Thomas splint
- Foot support
- Heel protectors

*Technique of skeletal traction:

Steps	1	2
1. Wash hands		
2. Explain procedure to the client		
3.Position client according to physician's request		
4.Physician performs skin preparation and discards materials in wastebasket		
5. Open pin/ wire tray using aseptic technique		

6. Maintain the position of the injured extremity as requested, while physician injects local anesthetic into sites desired. Nurse and other assistants support client,

limb, other tissues to be placed in traction

7.Assist as requested with application of Thomas splint,

Pearson attachment, spreader bar, foot support
8. Assist as requested with the positioning of the

affected extremity in suspension and in applying the

ropes, pulleys
and prescribed weights
9. Position the patient's body in proper alignment

10.Assess client's initial reaction or response to traction

before physician leaves
11. Gather equipment and supplies and return to proper

storage places
12.Raise side rails if appropriate

13. Wash hands

Note: https://youtu.be/E48BdevYnTI

Follow up activities

Complete the following:

1.List five principles of effective traction:

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2.Differentiate between skin and skeletal traction?

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3.Define cast?

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4.Discuss cast care?

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5. Choose the correct answer:

1. An appropriate technique for the nurse to implement for a client being placed in traction is to:
   a. Apply a traction boot tightly
   b. Drop the weights after the traction is attached
   c. Assess the neurovascular status every 1 to 2 hours for the first day
   d. Shave the hair off the area where the traction is to be placed
2. For a client who is to be placed in Buck traction the nurse prepares the:
   a. Occipital area
   b. Arm and forearm
   c. Back and abdomen
   d. Lower extremities

Module

*Module Unite: Burn wound care

*Module Title: Burn wound care

*Time Allotted: 1 hour theory and 30 Minute clinical practice

(Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with burn.
- Implements standardized protocol and guidelines when providing nursing care for patients with burn.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with burn.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with burn.
- Develop a teaching plan for patient with burn.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with burn.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Burn wound care

Outlines

1. Hydrotherapy
   - Definition of hydrotherapy,
   - Purpose of hydrotherapy
   - Procedure of hydrotherapy.
2. Debridement
   - Definition of debridement
   - Purposes of debridement
   - Types of debridement.
3. Burn wound dressing:
   - Types of wound dressing

Burn wound care

Removal of previous dressing

- Dressings that adhere to the wound can be removed more comfortably and without damaging healing tissue by moistening the wound with tap water.
- The remaining dressings are carefully and gently removed.
- The patient may participate in removing the dressings, providing some degree of control over this painful procedure.
- The wounds are then cleaned and debrided to remove any remaining topical agent, exudates, and dead skin. Sterile scissors and forceps may be used to trim loose eschar and encourage separation of devitalized skin.
- During this procedure, the wound and surrounding skin are carefully inspected. The color, odor, size, exudates, signs of re epithelialization, and other characteristics of the wound and the eschar and any changes from the previous dressing change are noted.

Wound Cleaning:

Various measures are used to clean the burn wound, such as hydrotherapy.

Hydrotherapy:

Hydrotherapy treatment involves the use of warm, running water to help the healing process of a burn injury by reducing infection and relieving pain.

Purpose:

- Removal of old dressing.
- Clean the wound through removal of non-viable tissue.
- Assessing the wound.
- Provide range of motion exercise and patient comfort.
- As body bathing.
- Promote wound healing.

Steps

1.If the patient is ambulatory, the wounds can be cleansed in a shower.

2.The wounds of non-ambulatory patients can be cleansed using shower carts—mobile stretchers made with removable sides, drainage holes, and positioning capabilities.

3.Retractable shower hoses suspended from walls and ceilings provide the nurse with easy access to a water source for washing the wounds.

4.Unstable patients may have their wounds washed at the bedside.

5.Total immersion hydrotherapy is rarely performed. Because of the high risk for infection and sepsis

6. The use of plastic liners, water filters, and thorough decontamination of hydrotherapy equipment and wound care areas is required to prevent cross-contamination.

7. The temperature of the water is maintained at 37.8C (100F), and the temperature of the room should be maintained between 26.6C and 29.4C (80F to 85F).

8. The duration of wound cleansing and dressing change is determined by the patient's ability to tolerate the treatment and to maintain a satisfactory body temperature.

9. During the bath, the patient is encouraged to be as active as possible. Hydrotherapy provides an excellent opportunity for exercising the extremities and cleaning the entire body.

10. When the patient is removed, any residue adhering to the body is washed away with a clear water spray or shower.

11. Unburned areas, including the hair, must be washed regularly as well.

12. At the time of wound cleaning, all skin is inspected for any hints of redness, breakdown, or local infection.

13. Hair in and around the burn area, except the eyebrows, should be clipped short or shaved.

14. Intact blisters should be left alone and debrided only if they rupture or break.

15. When nonviable loose skin is removed, aseptic conditions must be established.

16. Wound cleansing is usually performed daily in wound areas that are not undergoing surgical intervention.

17. Mechanical debridement can be performed to remove loose nonviable tissue.

18. After the burn wounds are cleaned, they are gently patted dry, and the prescribed method of wound care is performed.

19. Patient comfort and ability to participate in the prescribed treatment are also important considerations.

20. During the treatment, the patient is assessed for signs of chilling, fatigue, changes in hemodynamic status, and pain unrelieved by analgesic medications or relaxation technique.

Wound Debridement

The goals of, debridement are:

- Removal of tissue contaminated by bacteria and foreign bodies, thereby protecting the patient from invasion of bacteria
- Removal of devitalized tissue or burn eschar in preparation for grafting and wound healing

Types of debridement:

1. Natural Debridement or Autolytic Debridement:

This uses the body's own enzymes and moisture to re-hydrate, soften, and liquefy non-viable tissue. With natural debridement, the dead tissue separates from the underlying viable tissue spontaneously. Bacteria that are present at the interface of the burned tissue liquefy the fibrils of collagen that hold in place for the first or second post burn weeks. However, use of antibacterial topical agents tends to slow this natural process of eschar separation and slows the healing process.

2. Mechanical Debridement:

Mechanical debridement involves the use of instrument; surgical scissors, scalpels, and forceps to separate and remove the eschar. This technique can be performed by skilled physicians, nurses, or physical therapists and is usually done with daily dressing changes.

If bleeding occurs, pressure can be used to stop the bleeding from small vessels. Wet-to-dry dressings are not advocated in burn care because of the chance of removing viable cells along with necrotic tissue. Dressing changes alone aid the removal of wound debris.

3. Chemical Debridement or Enzymatic Debridement:

Chemical enzymes, derived from microorganisms including clostridium, are used to slough off necrotic tissue such agents usually do not have antimicrobial properties; they should be used together with topical antibacterial therapy to protect the patient from bacterial invasion.

Heavy metals such as silver deactivate the debriding agent; therefore, caution is necessary to ensure that the debriding agent does not interfere with the topical antimicrobial agent. Separate dressings are used to prevent this from occurring.

4. Surgical Debridement

Surgical debridement is an operative procedure involving either primary excision (surgical removal of tissue) of the full thickness of the skin down to the fascia (tangential excision) or shaving of the burned skin layers gradually down to freely bleeding, viable tissue. Surgical excision is initiated early in burn wound management. This may be performed within the first few days after the burn or as soon the patient is hemodynamically stable and edema has decreased.

- Before the procedure skin graft, if needed, and an occlusive dressing.
- If the wound bed is not ready for a skin graft at the time of excision, a temporary biologic dressing may be used until a skin graft can be applied during subsequent surgery
- The use of surgical excision carries with it risks and complications, especially with large burns. The procedure creates a high risk of extensive blood loss (as much as 100 to 125 mL of blood per percentage of body surface excised) and lengthy operating and anesthesia times.

Wound Dressing:

Purpose:

- Protect the wound from physical damage and micro-organisms
- Be comfortable, compliant and durable
- Be non-toxic, non-adherent, and non-irritant
- Allow gaseous exchange
- Allow high humidity at the wound
- Be compatible with topical therapeutic agents
- Be able to allow maximum activity for the wound to heal without retarding or inhibiting any stage of the process.

Methods of wound dressing

1- Open dressing: The open method is a popular form of treatment for burn wound; it involves application of topical agents without the use of dressing.

Advantages of open dressing:

- Continuous observation and assessment of
- wound Reduce possible hyperthermia
- Facilitate physical therapy
- Helpful in treating burn of face, neck, ear and perineum

Disadvantages of open dressing

- Hypothermia
- Require frequent linen and ointment
- Can't be used effectively in children and unconscious patient

2- Closed method: It involves the application of occlusive dressing over the burn wound.

Occlusive dressings:

- May be used over areas with new skin grafts to protect the graft and promote an optimal condition for its adherence to the recipient site.
- An occlusive dressing is thin gauze that is impregnated with a topical antimicrobial agent or is applied after application of a topical antimicrobial agent.
- Ideally, these dressings remain in place for 3 to 5 days, at which time they are removed for examination of the graft.
- When occlusive dressings are applied, precautions are taken to prevent two body surfaces from touching, such as fingers or toes, ear and scalp, the areas under the breasts, any point of flexion, or between the genital folds.
- Functional body alignment positions are maintained by using splints or by regular repositioning of the patient.

NURSING ALERT: Dressings impede circulation if they are too tightly wrapped. The peripheral pulses must be checked frequently and burned extremities elevated on two pillows. Extremities are always wrapped from distal to proximal to the heart. If the patient's pulse is diminished, this is a critical situation and must be addressed immediately.

Procedure

SUPPLIES

- Clean gloves
- Clean scissors

- Liquid antibacterial soap
- Water (bottled sterile water or saline, or from hose in hydrotherapy room)
- Sterile container
- Sterile field
- Sterile gloves
- Gauze (4 x 4s)
- Antimicrobial cream/ointment as ordered
- Mesh rolls gauze (large and/or small rolls)

Equipment for dressing: -

- Sterile fine mesh gauze
- I% silver sultfadiazine (silvadene)
- Antibiotic ointment (Neoporin.
- polysporin) Non adhering dressing
- Transparent wound dressing

Procedure:

Steps	Rationale
A-Prepare Patient:	
1. Explain the procedure to the patient	To reduce fear and gain cooperation
2. The patient adequately premeditated with analgesics as ordered (20 to 30 minutes before the procedure).	To decrease the pain

B-prepare Environment:

As discussed before in hydrotherapy

C-prepare The Nurse	
1. The nurse perform wound care should be aware with principles of infection control include:	To prevent infection
The use of hair cap, mask, sterile gown and gloves	
As well as practice sterile technique during all procedure	To facilitate handling
2. All supplies will be set up using sterile technique.	
D- Transport/ambulate patient to hydrotherapy room as applicable.	
E- clean the wound as discussed before in hydrotherapy	To prevent contracture and joint stiffness
F-Remove the old dressing if present	
G- Physical therapy exercise is performed with qualified person at the end of hydrotherapy	
II- Once the dressings are removed. The patent's burn wound are gently cleansed with gauze pad and then inspected.	
I-When washing wounds, the nurse use firm circular motion to cleanse cream, ointment, loose skin	
	To decrease bacteria

present

j- Any loose skin or eschar can be removed with gauze or debrided with scissors and forceps

K- The patient's unburned areas are then washed.

L- Wound dressing After wound cleaning, the burned areas are patted dry and the prescribed topical agent is applied

the wound is then covered with several layers of dressings.

A light dressing is used over joint areas to allow for motion (unless the particular area has a graft and motion is contraindicated). A light dressing is also applied over areas for which a splint has been designed to conform to the body contour for proper positioning.

Circumferential dressings should be applied distally to proximally.

If the hand or foot is burned, the fingers and toes should be wrapped individually to promote adequate healing. Initially fingers which have circumferential burns should be dressed with the finger tips exposed to monitor neurovascular status. Once

surround skin

To prevent contracture and joint stiffness

– Burns to the face may be left open to air once they have been cleaned and the topical agent has been applied. – Careful attention must be given to ensure that the topical agent does not interfere with the eyes or mouth. A light dressing can be applied to the face to absorb excess exudates that might run into the eyes, causing irritation. N- Record any observations in the patients wound and ask the doctor about any abnormal founding	

Note:https://youtu.be/l_jbLugZfBk

Follow Up Activities

Choose the correct answer:

1. The patient adequately premeditated with analgesics for about
   a. 1 to 2 hour sbefor ethe procedure
   b. 20 to 30 minutes before the procedure
   c. From 15 to 30 minutes after the procedure
   d. 10 to15 minutes befor ethe procedure

2. Which type of debridement is the best for burns of large body surface area
   a. Natural.
   b. Chemical.
   c. Surgical.
   d. Mechanical .

3. Which type of dressing as a technique used as a mechanical debridement
   a. Dry sterile dressing.
   b. Wet to dry sterile dressing.
   c. Moist transparent wound barrier.

Complete the following:

1. Define hydrotherapy:

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2. What is the purpose of hydrotherapy?

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3. List types of debridement?

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Module

Module Unit: - Neurological Procedures

Module Title: - Lumbar Puncture

Time Allocated: - one-hour theory / 30 Minute clinical practice

(Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with cast.
- Implements standardized protocol and guidelines when providing nursing care for patients with cast.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with cast.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with cast.
- Develop a teaching plan for patient with cast.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with cast.

Teaching and Learning Activities:

a)Oral Presentation.

b) Group participation

c)Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

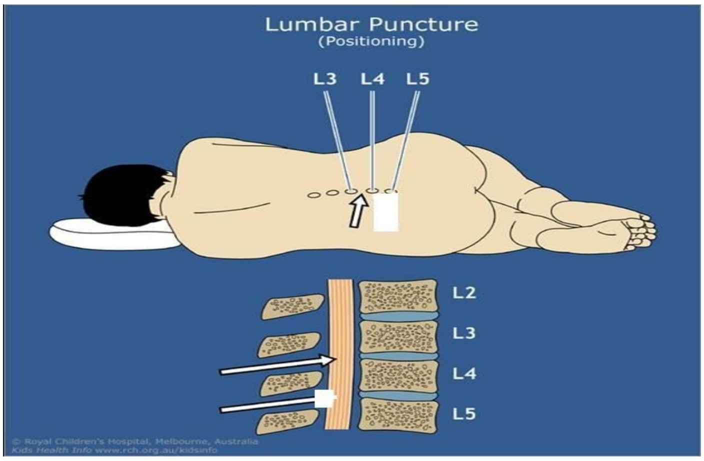
Outlines:

1. Definition.
2. Purpose.
3. Indications.
4. Contraindications.
5. Equipment.
6. Procedure.

Lumbar Puncture

Definition:

Lumbar puncture (LP), also known as a spinal tap, is a medical and invasive procedure in which a hollow needle is inserted into lumbar subarachnoid space, usually between the third and fourth lumbar vertebra, and cerebrospinal fluid is withdrawn for diagnostic and therapeutic purposes.



Purpose:

1- Obtaining cerebrospinal fluid for examination.

2- Measuring cerebrospinal pressure and assisting in detection of obstruction of cerebrospinal fluid circulation.

3- Determining the presence or absence of blood in the spinal fluid.

4- Administering antibiotics and cancer chemotherapy intra ethically

5- Aiding in the diagnosis of viral or bacterial meningitis, subarachnoid or intracranial hemorrhage, tumors, and brain abscesses. Indications

●Symptoms of fever, malaise, and central nervous system irritability

Contraindications:

1.Lumbar skin infection

2.Raised intracranial pressure.

3.Suspected spinal cord mass or intracranial mass lesion.

4.Platelet count less than 50,000/μL

5.Spinal column deformities.

6.Lack of patient cooperation.

<u>Equipment:</u>

Sterile lumbar puncture set or:

1-Spinal Needle, 20 and 22-gauge 2-Manometer

3- Three-way stopcock

4- Sterile drapes

5- Xylocaine 1%- 2%.

6- skin antiseptic

7- Adhesive dressing

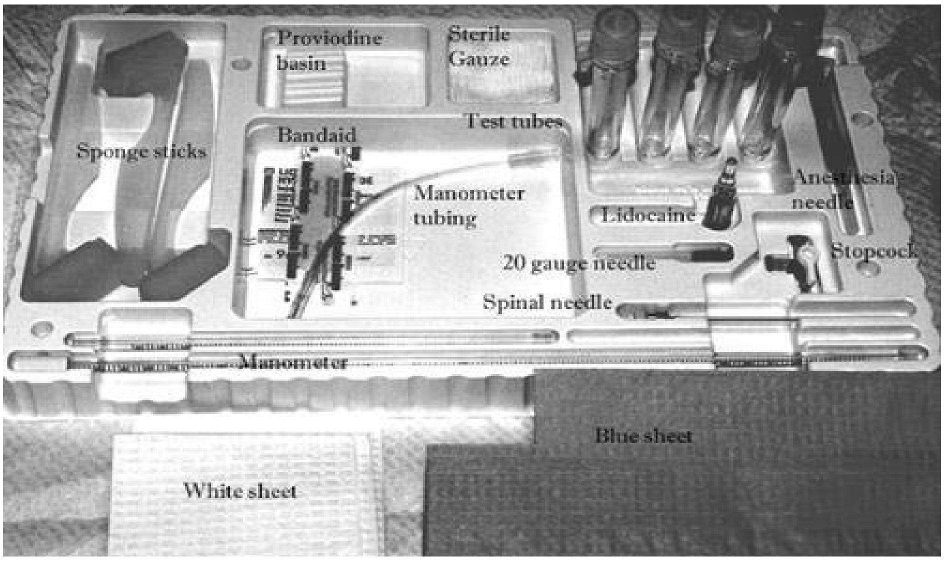
8- Sponges - 10 X 10 cm

9- Sterile gloves.

10- CSF specimen collection bottles

11- 22- to 23-gauge needle

12 - Mask



Procedure of lumbar puncture

Nursing action	Rationale
Preparatory phase:	
1- Ask the patient to empty urinary bladder and bowel.	1- To enhance comfort.
2- Explain procedure to patient.	2- Reassures the patient and gains cooperation.
3- If patient has signs of increased intracranial pressure CT should be done to rule out mass effect.	3- Removal of cerebrospinal fluid (CSF) in the presence of mass effect may precipitate brain herniation.
4- Positioning:	
a- Position the patient laterally with a small pillow under head and a pillow between legs. Patient should be lying on a firm surface.	a- The spine is maintained in a horizontal position. The pillow between the legs prevents the upper leg from rolling forward.
b- Ask patient to arch lumbar segment of back and draw knees up to abdomen, chin to chest, clasping knees with hands.	b- This posture offers maximal widening of the interspinous spaces and
c- Assist patient in maintaining this position by supporting behind knees	

and

affords easier entry into the

neck.

5- Alternately, for sitting position, have the patient straddle a straight- back chair (facing the back) and rest head against arms, which are folded on the back of the chair

<u>Performance phase:</u>

Skin is prepared with antiseptic solution and the skin

- The subcutaneous spaces are infiltrated with local anesthetic agent.

2- A spinal puncture needle is introduced at L3-L4 interspaces.

3- After the needle enters the subarachnoid

subarachnoid space.

c- Supporting the patient helps prevent sudden movement, which can produce a traumatic tap and thus impede correct diagnosis.

5- In obese patient and those who have difficulty in assuming an arched side- lying position, this posture may allow more accurate identification of the spinous processes and interspaces.

1- To reduce risk of contamination pain.

2- This area has enough amount of cerebrospinal fluid.

3-This maneuver prevents a false

increase in intraspinal pressure. Muscle space, help the patient to slowly straighten legs.

4- Instruct the patient to breathe quietly.

5- Ask the patient to not talk.

6- The initial pressure reading (opening pressure) is obtained by measuring the level of the fluid column after it comes to rest.

7- About 2-3 ml spinal fluid is placed in each of three test tubes

8- Closing pressure is the pressure on the manometer after the cerebrospinal fluid specimen is collected and / or excess cerebrospinal fluid removed.

<u>Follow-up phase:</u>

1- After procedure, the patient is instructed to remain in prone position without pillow for about 2 hours.

2- Ensure adequate hydration with oral or parental fluids.

tension and compression of the abdomen give falsely high pressures.

4- Hyperventilation may lower a truly elevated cerebrospinal fluid pressure.

5- Talking can elevate cerebrospinal fluid pressure.

6- With respiration, there is normally some fluctuation of spinal fluid in the manometer. Opening pressure exceed normal range when hydrocephalus or increased intracranial pressure is present. 7- for observation, comparison, and laboratory analysis. If traumatic puncture, blood should gradually clear with subsequent specimens.

8- Closing pressure should be within normal cerebrospinal fluid pressure range.

1- This reduces pressure on tissue surfaces along the needle track to prevent cerebrospinal fluid leakage.

2- This facilitates replacement of cerebrospinal fluid and prevents spinal headache.

3- Monitor for spinal headache, and observe for cerebrospinal fluid leak.	3- Cerebrospinal fluid leaks from lumbar punctures are characterized by intractable spinal headache.

Complication:

1- Severe headache.
2- Herniation of brain due to sudden decrease in pressure.
3- Accidental puncture of spinal cord
4- Meningitis from introducing bacteria into CSF.
5- Temporary voiding problems.
6- Slight elevation of temperature.
7- Back or leg pain/paresthesia.

Prevention of post lumbar headache:

A- Use small gauge needle.

B- Keep patient in prone position after procedure.

C- If more than 20 ml CSF removed patient should positioned in prone position for 2 hours, then flat in a side lying position for 2-3 hours, then in prone or supine for 6 more hours.

N.B.

Signs of increase intracranial pressure:

a- Early signs:

1) Disorientation, restlessness, increases respiratory effort, mental confusion.
2) Papillary change.
3) Weakness in one extremity or one side of the body.
4) Headache.

b- Late signs:

1- Level of consciousness continues to deteriorate may lead to coma.
2- Decrease in pulse and respiratory.
3- Increase in blood pressure and temperature.
4- Projectile vomiting.
5- Hemiplegia.

Characteristic of normal cerebrospinal fluid:

1- Normal range of spinal fluid pressure 50-175 mm H2o
2- clear and colorless appearance.
3- Ph 7.30 to 7.40
4- Fasting glucose 40 to 80mg/dl
5- White blood cells 0 to 5 small lymphocytes/mm3
6- Bloody spinal fluid may indicate subarachnoid hemorrhage or traumatic puncture.

Note:https://youtu.be/5vejEe-ShlI

Follow up activities

Choose the best answer for the following:

1- Which of the following is not a complication of lumbar puncture?

a. Epidermoid tumor implantation

b. Olfactory disturbances

c. Post lumbar puncture headache

d. Spinal column deformities

e. Local back pain

2- The position of a patient for lumbar puncture should be

a. Back arched out with hips and shoulders perpendicular to the bed

b. Back straight with hips and shoulders perpendicular to the bed.

c. Back arched out with shoulders rolled towards the bed and hips perpendicular to the bed.

d. Back straight with shoulders rolled towards the bed and hips perpendicular to the bed

e. Back not arched out with shoulders rolled towards the bed and hips perpendicular to the bed.

Complete the following:

1) List 2 contraindications of lumbar puncture:

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2) List 2 indications of lumbar puncture:

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3) List 2 complications of lumbar puncture:

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4) List 2 prevention of post lumbar headache:

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5) List 2 nurse role in the lumbar puncture procedure:

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<u>Module</u>

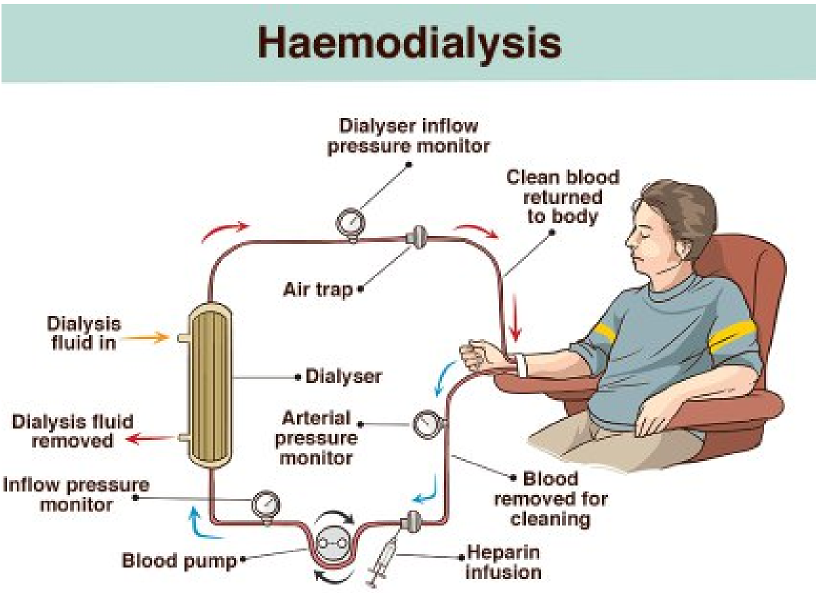
Module Unit: - Renal Procedures

Module Title: - Renal dialysis

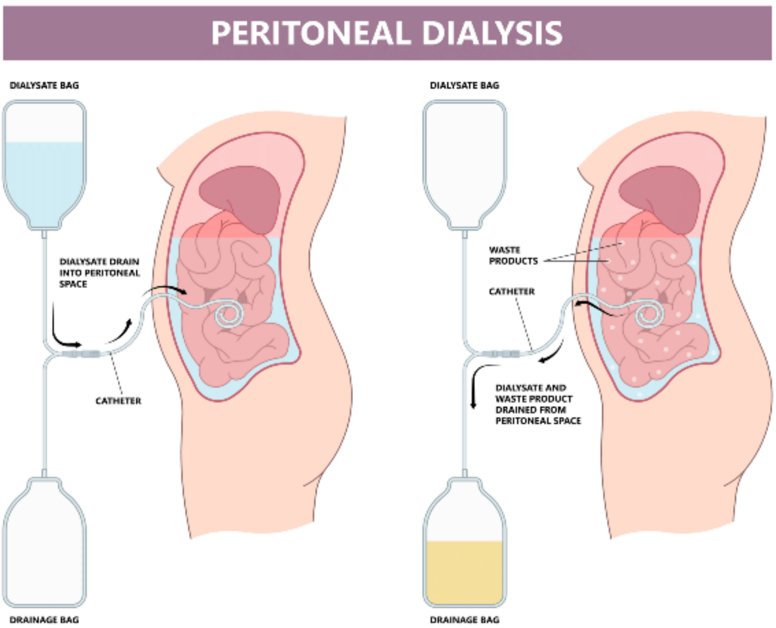
<u>Definition</u>: a life-sustaining treatment for end-stage kidney failure that removes waste, toxins, and excess fluid from the blood when kidneys can no longer function

Types of Dialysis:

- [Hemodialysis]: Blood is pumped out of the body, filtered through a dialyzer (artificial kidney) machine, and returned. Usually done 3 times a week for 3–5 hours per session.



- [Peritoneal Dialysis]: A cleaning solution (dialysate) is instilled into the abdomen via a catheter, allowing waste to be filtered directly inside the body.



Indications of hemodialysis:

- Acute kidney injury
- Uremic encephalopathy
- Pericarditis
- Life-threatening hyperkalemia
- Hypervolemia causing end-organ complications (e.g., pulmonary edema)

Contraindications of hemodialysis:

Absolute contraindication to hemodialysis is the inability to secure vascular access, and relative contraindications include:

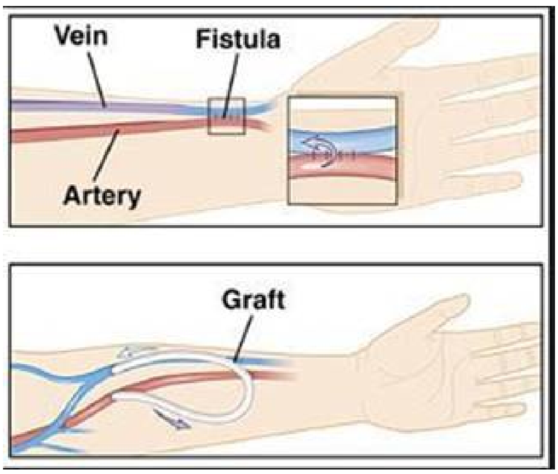
- Difficult vascular access
- Needle phobia
- Cardiac failure
- Coagulopathy

Complications

The most common complications associated with hemodialysis are:

- Intradialytic hypotension
- Dialysis disequilibrium syndrome: This is more common in patients during or soon after their first treatment. It is a clinical syndrome characterized by neurologic deterioration, restlessness, mental confusion, headache, occasional muscle twitching, and coma.
- Hemolysis
- Air embolism

Nursing action	Rationale
Pre hemodialysis session	
Assess vital signs	To establish a "hemodynamic baseline" and ensure the patient is stable enough to tolerate the stress of fluid removal
Auscultate lung sounds	To detect pulmonary congestion and guide fluid removal (ultrafiltration)
Monitor lab investigations as blood urea, creatinine, sodium, potassium, calcium, albumin and hemoglobin.	to evaluate the severity of uremic toxicity and electrolyte imbalances, albumin marker for nutritional status and "oncotic pressure , hemoglobin t o assess for anemia which is common in renal patients.
Check for holding any	To prevent severe intradialytic hypotension

antihypertensive medications	
Observe any inflammation signs as: √ Localized erythema √ Swelling √ Tenderness √ Pain in the access arm	To detect early signs of infection or thrombosis to prevent systemic sepsis and preserve the life-sustaining vascular access
Observe site of access for any abnormalities as: √ Vascular swelling √ Redness around puncture site √ Aneurysm	To ensure the integrity and patency of the vascular access to prevent life-threatening complications and ensure a successful treatment
Prepare hemodialysis machine, dialyzer, dialysate and supplies	
During hemodialysis session	
<u>Following aseptic technique during cannulation procedure as</u> Perform hand hygiene Wearing PPE Appling skin antiseptic solution appropriately Allow skin antiseptic solution to dry No contact with fistula/ graft's site (after disinfection) Perform cannulation aseptically by avoiding touch needle Cover puncture site with sterile dressing Insert needles in vascular access Remove gloves Perform hand hygiene	 

Assess vital signs especially blood pressure every 30 minutes	To detect and manage intradialytic hypotension early, and ensuring patient safety during rapid fluid shifts
Observe any hemodialysis complications as: Hypotension Hypertension Hypoglycemia Hyperglycemia Leg cramps Febrile reaction Dysrhythmia Bleeding at the access site Monitor patient for disequilibrium syndrome: √ Nausea √ Vomiting √ Restlessness √ Headache √ Seizures	
Observe Vascular access patency Uncovered during hemodialysis session Don't sleep on access arm Avoid using the patient's dialysis access arm for administering I.V. medications, taking BP, or blood sample	To ensure continuous blood flow and prevent clotting
Post- dialysis nursing care	
Following aseptic technique during decannulation procedure as: Hand hygiene Wearing PPE Remove needles aseptically and compress the site of insertion by sterile	

dressing Disinfect site of puncture Cover the site of puncture with sterile gauze or dressing Removing gloves Perform hand hygiene	
<u>Assess vital signs</u>	
Auscultate lung sounds	To confirm that the lungs are clear and the fluid removal was successful. And to ensure that patient is leaving the session without any remaining fluid congestion.
Monitor lab investigations as blood urea, creatinine, sodium, potassium, calcium, albumin and hemoglobin.	To confirm the treatment was effective and ensure that waste products and electrolytes have reached safe, stable levels
Instructions given to patients for vascular access maintenance as: √ Avoid wearing tight clothes at vascular access arm. √ Avoid carrying heavy weight at vascular access arm. √ Avoid measuring blood pressure from vascular access arm. √ Avoid laying on vascular access arm. √ Others	
Discard all equipment and supplies in special places	

Module Unit: - Renal Procedures

Module Title: - Renal Biopsy

Time Allocated: - one-hour theory / 30 Minute clinical practice (Demonstration and return demonstration)

Objectives:

- Integrate ethical, legal, sociocultural, and professional standard when providing care to patients with renal biopsy.
- Implements standardized protocol and guidelines when providing nursing care for patients with renal biopsy.
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patients with renal biopsy.
- Describe the responsibilities of the nurse in meeting the psychological needs of the patients with renal biopsy.

- Develop a teaching plan for patient with renal biopsy.
- Manage time effectively and set priorities.
- Conducts appropriate nursing activities skillfully and in accordance with best evidence-based practice.
- Communicate effectively with all staff members in inter- professional context to improve patient's outcomes.
- Convey a positive attitude toward other team members while working with patients with renal biopsy.

Teaching and Learning Activities:

a) Oral Presentation.

b) Group participation

c) Demonstration and return demonstration of colostomy care.

d) Using videos related topic.

Outlines:

1. Types of renal biopsy.

2. Definition of percutaneous renal biopsy.

3.Indications of Renal Biopsy.

4.Contraindications of Renal Biopsy.

5.Equipment.

6.Renal Biopsy Preparation.

7.Renal Biopsy Procedure.

8.Post Renal Biopsy Care.

9.Renal Biopsy Complications.

Renal Biopsy Procedure

Types of renal biopsy:

- Percutaneous renal biopsy: A native renal biopsy is commonly performed percutaneously through the patient's back. The biopsy needle is typically guided using ultrasound.

●Transjugular kidney biopsy and laparoscopic/open kidney biopsy.

Definition of percutaneous renal biopsy:

- Percutaneous kidney biopsy (through the skin): is the removal of a sample of kidney tissue for diagnosis or to evaluate the function of a transplanted kidney and it is performed through the patient's back. The biopsy needle is typically guided using ultrasound.

Notes to be remembered:

A nephrologist (kidney and renal system specialist) performs the procedure on an outpatient basis, usually in the radiology department of the hospital. Preparation, biopsy, and recovery take several hours

Indications of Renal Biopsy:

- Unexplained renal failure
- Acute nephritic syndrome
- Nephrotic syndrome
- Non nephrotic proteinuria
- glomerular hematuria
- Renal masses (primary or secondary)
- Renal transplant rejection
- Renal transplant dysfunction

Contraindications of Renal Biopsy

The most common contraindication to kidney biopsy is:

- Bleeding disorder (e.g., inability to coagulate, platelet abnormality)
- Hypertension
- Pyelonephritis (disease of the pelvis of the kidney)
- Shrunken kidney (i.e., as a result of disease)
- Tumor.

Equipment:

- Marker pen
- Sterile field sheet
- Sterile fenestrated drapes

- Sterile gloves

- Syringe 10 cc
- Antiseptic solution (betadine)
- K y gel
- Sterile gauze and dressing
- sponge
- forceps
- Adhesive tape.
- Kidney basin
- Needle set: usually an 18 G (core biopsy needle)
- 1% lidocaine/lignocaine
- Midazolam (for sedation): select cases only; assess on a case-by- case basis
- Histopathology department pots

Renal Biopsy Preparation

- Renal imaging: Two normal size, unscarred, unobstructed kidneys.
- Measure blood pressure: diastolic blood pressure >95 mmHg
- Urine culture: Sterile (Indicating absence of infection)
- Assess coagulation status:
- The following medications must be discontinued before biopsy to reduce bleeding:

.Aspirin (10 days before)

2.Warfarin (Coumadin, Jantoven, generics), Heparin, and other anticoagulants

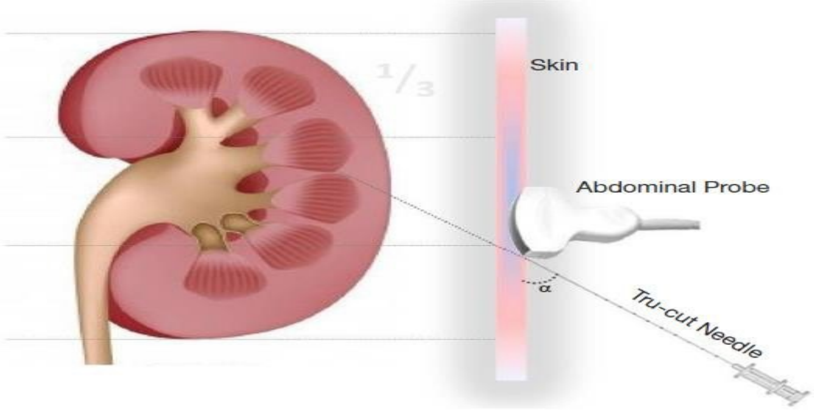
.Nonsteroidal anti-inflammatory drugs (e.g., ibuprofen)

- Platelet count <100 x 10^3/ml

1. Typically, blood platelet count is evaluated to ensure that the blood will clot properly after the procedure.

2. Low blood platelet count may indicate that the patient has a disorder that makes a biopsy inadvisable.

- Monitor activated partial thromboplastin time (APTT) and bleeding time.
- Avoid eating or drinking for eight hours before the procedure.
- Ask patient to sign the informed consent of the procedure.



Renal Biopsy Procedure:

- Make sure that the informed consent is signed.
- Biopsy is performed on one kidney.
- The patient lies on his or her stomach (Prone position) on a firm table with cushioning under the abdomen for support and must not move during the procedure.
- The kidney is located with ultrasound or CT scan.
- The biopsy site is marked over the right kidney
- Light sedation may be needed.
- Local anesthetic is injected (1-2% Lignocaine from the skin down to the capsule)
- Once the area is numb, the nephrologist uses ultrasound to view the kidneys in real time and guides a spring-loaded biopsy needle into the kidney.
- The needle removes about 5 to 15 glomeruli.

- Two or three samples are usually required, each the size and shape of one-half inch of string.
- After the core tissue samples are extracted, pressure is applied to the incision for 2-5 minutes to slow bleeding, and a bandage is placed on the wound.
- Document procedure (Date, time, site, observations and signature).

Fig. 2. Prone native kidney biopsy position. A posterior approach is used for native kidneys. The wedge is placed under the abdomen to eliminate the lumbar lordosis.

Post Renal Biopsy Care

- Enforce bed rest for the patient until stable:
  - The patient must lie on his or her back for 8 to 24 hours.
  - Some nephrologists advocate a 24-hour recovery in the hospital, where the patient can be observed for complications.
- Monitor vital signs (heart rate and rhythm, blood pressure, respiratory status, and body temperature):
  - Every 15 mins for 1 hour.
  - Every 30 mins for 1 hour.
  - Every hour for 4 hours.
  - Every 4 Hours for net remaining 24 hours.
- Monitor for bleeding from the insertion site or biopsy site:-

- The kidney contains many blood vessels and bleeds as a result of the biopsy.
  - Bright red (arterial) blood in the urine (hematuria) may be seen for the first 24 hours.
  - If blood is seen in the urine after 24 hours, further care may be required to stop the bleeding.
- Save aliquots of each voided urine sample in clear specimen jars (Test urine for presence of blood).
- Hct monitored 6-8 hours and 18-24 hours after biopsy.
- Many people experience muscle aches and general soreness during recovery.
- Report any problems, such as:
  - Bloody urine for more than 24 hours after the biopsy.
  - Unable to pass urine.
  - Fever.
  - Worsening pain at the biopsy site.
  - Feeling faint or dizzy.

Renal Biopsy Complications

The following complications may occur:

- Infection
- Acute blood loss (in as many as 10%)
- Bleeding requiring surgery, transfusion, or kidney removal (less than 1%)
- Death (less than 1%)
- Pain lasting longer than 12 hours (in 4%)

Note:https://youtu.be/kIqTBW3Vcc8

Follow Up Activities

Choose the correct answer:

1. Indications of renal biopsy include all of the following <u>except:</u>

a) Acute nephritic syndrome

b) Nephrotic syndrome

c) Renal transplant rejection

d) Increased ICP

Complete the following:

1.Contraindications of renal biopsy are:

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2.Renal Biopsy Complications are:

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3. Give notes about post renal biopsy care :

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